



Complete Product & System Documentation

Version 2.4.1 | Generated April 2026 | DeeX Technologies Ltd

1. Product Overview

DeeX Product Overview

Version: 2.4.1 | **Build:** #1842

Organization: DeeX Technologies Ltd (DEEX OPTIONS LTD)

Document Date: April 2, 2026

Status: Frontend Prototype — Investor & Engineering Reference

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1. Executive Summary

DeeX is a Nigerian fintech platform that combines cryptocurrency trading, gift card exchange, bill payments, peer-to-peer transfers, and merchant payment processing into a single mobile-first application. Built to serve the Nigerian market with NGN-denominated transactions, DeeX bridges the gap between traditional finance and digital assets while providing a comprehensive suite of everyday financial services.

This repository contains the **frontend prototype** — a fully interactive, mobile-first React application that demonstrates the complete user experience. A separate backend system handles real transaction processing, wallet management, and compliance workflows.

2. Product Vision

To become Nigeria's most trusted all-in-one digital finance platform — where users can trade crypto, sell gift cards, pay bills, send money, and manage virtual cards through a single, intuitive interface.

Guiding Principles

- **Accessibility:** Serve both crypto-native users and first-time digital finance users
- **Speed:** Near-instant transactions with minimal friction
- **Trust:** Multi-tier KYC, audit trails, and transparent fee structures

- **Rewards:** Gamified loyalty system that incentivizes engagement and referrals

3. Target Market

Primary Market: Nigeria

Parameter	Detail
Currency	Nigerian Naira (NGN)
Identity Verification	BVN (Bank Verification Number)
Banking	Nigerian commercial banks (access bank, GTBank, etc.)
Telecom	MTN, Glo, Airtel, 9mobile
Electricity	IKEDC, EKEDC, and other disco providers
Betting	Bet9ja, SportyBet

Market Opportunity

- Nigeria has one of the largest crypto adoption rates globally
- High demand for alternative remittance and payment rails
- Growing gift card resale market
- Underserved retail investor segment seeking easy crypto access

4. User Personas

4.1 Retail User (Primary)

- **Profile:** 18–45 years old, smartphone owner, banked or underbanked
- **Goals:** Buy/sell crypto at competitive rates, sell unused gift cards, pay bills quickly, send money to friends/family
- **Pain Points:** High exchange rates, slow bank transfers, fragmented apps for different services
- **Key Features Used:** Crypto trading, gift card trading, bill payments, DeeX Pay, rewards

4.2 Merchant

- **Profile:** Small to medium business accepting digital payments
- **Goals:** Receive payments from customers, manage payouts, track transaction history
- **Key Features Used:** DeeX Pay, payment links, payout management

4.3 Admin / Operations

- **Profile:** Internal DeeX staff (e.g., Adedamola A., Dawood K.)
- **Goals:** Monitor platform health, approve KYC/rejections, manage rates, handle compliance alerts, oversee payouts
- **Key Features Used:** Admin dashboard, KYC review, compliance engine, audit logs, report generation

4.4 Influencer (Referral Partner)

- **Profile:** Social media influencers, community leaders
- **Goals:** Earn through referral profit-share, grow their audience's adoption
- **Key Features Used:** Referral dashboard, influencer profit tracking, leaderboard

5. Core Value Propositions

Proposition	Description
All-in-One Finance	Crypto, gift cards, bills, transfers, virtual cards — one app
Competitive NGN Rates	Real-time rates for crypto and gift cards denominated in Naira
Rewards That Matter	DeeXPoints (1pt = ₦10), cashback, trade streaks, referral bonuses
Trust & Compliance	3-tier KYC, BVN verification, 2FA, audit logging, compliance monitoring
Speed & Simplicity	Mobile-first design, PIN-confirmed transactions, instant receipts
Merchant Enablement	Payment links, merchant onboarding, seamless payout processing

6. Feature Summary

6.1 User-Facing Features

Feature	Description	Status
Crypto Trading	Buy/sell BTC, ETH, USDT, USDC, SOL, TRX, DOGE with live NGN rates	Prototype
Crypto Swap	Swap between crypto assets (0.5% fee)	Prototype
Crypto Withdrawal	Send crypto to external wallets with PIN confirmation	Prototype
Crypto Deposit	Generate deposit addresses for supported assets	Prototype
Gift Card Trading	Sell gift cards (Apple, Google Play, Amazon, Steam, iTunes, Walmart, Nike, Sephora) with tiered NGN rates	Prototype
Bill Payments	Airtime, Data, Electricity, Betting across Nigerian providers	Prototype
DeeX Pay	Pay merchants, send cash to bank accounts, generate payment links	Prototype
Virtual Cards	Visa virtual cards (USD), create/fund/freeze/delete (KYC Level 2 required)	Prototype
Rewards System	DeeXPoints (1pt = ₦10), referral bonuses, trade streaks, cashback, redemption	Prototype
Referral Program	Earn 100pts when referral trades >\$100, leaderboard, influencer profit-share	Prototype
KYC Verification	3-tier system: Level 1 (BVN + liveness), Level 2 (ID + address), Level 3 (employment + source of income + risk questionnaire + 2FA)	Prototype
Security	PIN lock, biometric login, 2FA, session management, login history	Prototype
Notifications	Login alerts, session terminations, promotional messages	Prototype
Support	Live chat, email (support@deex.app), WhatsApp (+234 810 367 4006), FAQ	Prototype
Activity & Receipts	Transaction history, detailed receipts, asset-level breakdowns	Prototype
Profile Management	Edit profile, linked bank accounts, statement generation	Prototype

6.2 Admin Dashboard Features

Feature	Description	Status
Dashboard Overview	Metrics (total payout, active users, customers, merchants), performance charts	Prototype
Wallet Management	DeeX wallet + customer wallet management, auto-swap, auto-withdrawal, crypto assets, trade volume, asset distribution	Prototype
Order Management	Order tracking with auto-pay toggle, retry failed payouts	Prototype
Payout Tracking	Payout monitoring with status filters	Prototype
Virtual Card Admin	Card issuance monitoring, settings	Prototype
Gift Card Admin	Order approval/rejection, brand management, rate configuration, analytics	Prototype
Bill Payment Monitoring	Transaction monitoring by type and status	Prototype
User Management	Customer list with search, status filters, KYC levels	Prototype

KYC Review	Approval/rejection workflow, document review, limit management	Prototype
Compliance Engine	Alert management (high-frequency trading, multi-device, large withdrawal, failed KYC, wash trading), configurable rule thresholds	Prototype
Audit Log	All admin actions recorded with timestamps	Prototype
Rewards Admin	Points configuration, payout approval, earner rankings, influencer management	Prototype
Reports	Revenue, Growth, Retention, Website, Happiness, Survey reports	Prototype
Settings	Security, fees, payroll management, rewards configuration	Prototype

6.3 Route Map (Frontend Pages)

The application exposes 36 distinct routes:

Route	Page	Purpose
`/onboarding`	Onboarding	First-time user introduction
`/login`	Login	User authentication
`/pin`	PinLock	PIN verification
`/signup`	SignUp	New user registration
`/dashboard`	Dashboard	User home screen
`/wallet`	Wallet	Asset balances overview
`/deposit`	Deposit	Crypto deposit addresses
`/sell-crypto`	SellCrypto	Sell crypto for NGN
`/swap-crypto`	SwapCrypto	Crypto-to-crypto swap
`/withdraw`	WithdrawCrypto	Withdraw to external wallet
`/send-money`	SendMoney	P2P and bank transfers
`/asset/:symbol`	AssetDetail	Individual asset view
`/receipt`	Receipt	Transaction receipt
`/transaction-detail`	TransactionDetail	Transaction details
`/deex-pay`	DeexPay	Merchant payments & links
`/giftcards`	GiftCards	Gift card trading
`/bills/:type`	BillPayment	Bill payments by category
`/activity`	Activity	Transaction history
`/rewards`	Rewards	Points & rewards management
`/referrals`	ReferralDashboard	Referral tracking
`/notifications`	Notifications	Push/system notifications
`/profile`	Profile	User profile
`/edit-profile`	EditProfile	Profile editing

<code>`/generate-statement`</code>	GenerateStatement	Account statement export
<code>`/about`</code>	AboutDeeX	About DeeX information
<code>`/security`</code>	SecuritySettings	Security configuration
<code>`/bank-accounts`</code>	BankAccounts	Linked bank accounts
<code>`/kyc`</code>	KycVerification	KYC verification flow
<code>`/support`</code>	Support	Help & support
<code>`/quick-action`</code>	QuickAction	Quick action shortcuts
<code>`/virtual-cards`</code>	VirtualCards	Virtual card management
<code>`/trade-streak`</code>	TradeStreak	Trade streak gamification
<code>`/admin`</code>	AdminPanel	Admin dashboard
<code>`/admin/users/:id`</code>	AdminUserDetail	Individual user detail view

7. System Architecture & Classification

7.1 Repository Scope

This repository is a **frontend prototype** built with React + Vite + TypeScript. It demonstrates the complete user interface and interaction flows of the DeeX platform. All data is mocked using:

- ``localStorage`` for session state and user preferences
- Hardcoded arrays and objects in ``src/data/`` for business data
- Simulated API responses via TanStack Query mock patterns

7.2 Backend Separation

A separate backend system (not in this repository) is responsible for:

- Real transaction processing and settlement
- Wallet management and blockchain interactions
- KYC document storage and verification
- Payment gateway integrations (banks, telcos, billers)
- Compliance monitoring and reporting
- User authentication and authorization

- Rate feeds and pricing engines

7.3 Design Philosophy

- **Mobile-first:** User-facing screens are constrained to a 430px max-width container, simulating a native mobile app experience
- **Admin desktop:** The admin dashboard uses a responsive desktop layout with sidebar navigation
- **Component-driven:** Built with shadcn-ui primitives for consistency and accessibility
- **Animation-rich:** Framer Motion provides smooth transitions and micro-interactions

8. Technology Stack

8.1 Core Framework

Technology	Version	Purpose
React	18.3.1	UI framework
Vite	5.4.19	Build tool and dev server
TypeScript	5.8.3	Type-safe JavaScript
React Router	6.30.1	Client-side routing

8.2 UI & Styling

Technology	Purpose
Tailwind CSS	Utility-first CSS framework
shadcn-ui	Accessible component primitives (Radix UI based)
Lucide React	Icon library
Framer Motion	Animations and page transitions
class-variance-authority	Component variant management
next-themes	Dark/light theme support

8.3 Data & Forms

Technology	Purpose
TanStack Query (React Query)	Server state management and caching

React Hook Form	Form handling
Zod	Schema validation
Recharts	Data visualization and charts

8.4 Testing & Quality

Technology	Purpose
Vitest	Unit and component testing
Testing Library	React component testing utilities
ESLint	Code linting
TypeScript ESLint	TypeScript-specific linting rules

8.5 Additional Libraries

Technology	Purpose
date-fns	Date formatting and manipulation
Sonner	Toast notifications
vaul	Mobile drawer component
cmdk	Command palette / search
embla-carousel-react	Carousel/slider component
input-otp	OTP input component

9. Key Business Metrics

The following metrics are represented in the prototype's mock data and reflect target operational scale:

Metric	Value
Total Payout Volume	₦2,396,106,090.97
Active Users	1,847
Total Customers	2,833
Registered Merchants	12
Failed Payouts	364
Payout Success Rate	~84.3%

Points & Rewards Economics

Parameter	Value
DeeXPoint Value	1 point = ₦10

Referral Bonus	100 points (awarded when referral trades >\$100)
Swap Fee	0.5%

10. Assumptions & Limitations

10.1 Prototype Limitations

1. **No Real Transactions:** All trades, transfers, and payments are simulated. No blockchain interactions or bank API calls occur.
2. **Mocked Authentication:** Login/signup flows use localStorage — no real auth provider is connected.
3. **Static Data:** Rates, balances, and transaction histories are hardcoded and do not update in real time.
4. **No Persistence Across Sessions:** Data stored in localStorage is browser-specific and not shared.
5. **Single-User Simulation:** The app does not support multi-user scenarios or concurrent sessions.
6. **No File Uploads:** KYC document upload flows are UI-only — no actual file processing occurs.
7. **Admin Access Unrestricted:** The admin panel is accessible via route without role-based access control in the prototype.

10.2 Backend Assumptions

The following capabilities are assumed to exist in the separate backend system:

1. Real-time crypto rate feeds from liquidity providers
2. Blockchain node integration for deposits and withdrawals
3. Bank API integration for NGN settlements (NIP, NIBSS)
4. BVN verification via licensed identity provider
5. Gift card validation and redemption pipeline
6. Bill payment provider integrations (telcos, discos, betting platforms)
7. Virtual card issuance via card program manager
8. Push notification service (FCM/APNs)
9. Email and SMS delivery infrastructure
10. Document storage and KYC processing pipeline
11. Fraud detection and AML monitoring
12. Audit trail and regulatory reporting

10.3 Regulatory Assumptions

1. DeeX Technologies Ltd holds or is pursuing necessary CBN and SEC licenses for crypto asset services in Nigeria
2. BVN verification complies with NIBSS requirements
3. KYC tiers align with CBN tiered KYC guidelines
4. Virtual card program complies with Visa network rules
5. AML/CFT policies are implemented in the backend compliance engine

11. Roadmap Considerations

11.1 Frontend-to-Backend Integration

Priority tasks for connecting this prototype to the production backend:

1. Replace mock data with API calls (TanStack Query is already configured for this pattern)
2. Implement real authentication flow (JWT/session tokens)
3. Add role-based route guards for admin panel

- 4. Connect WebSocket or SSE for real-time rate updates
- 5. Implement proper error boundaries and loading states

11.2 Feature Enhancements

- 1. **Portfolio Analytics:** Add portfolio performance charts, P&L tracking
- 2. **Recurring Payments:** Scheduled bill payments and auto-buy for crypto
- 3. **Multi-Currency Support:** Expand beyond NGN to include GHS, KES for regional expansion
- 4. **P2P Marketplace:** Direct user-to-user crypto trading with escrow
- 5. **Savings Vaults:** Locked savings products with yield
- 6. **API for Merchants:** Developer-facing API for DeeX Pay integration

11.3 Platform Expansion

- 1. React Native or Capacitor wrapper for native mobile app distribution
- 2. Progressive Web App (PWA) capabilities for offline access
- 3. Desktop admin app with Electron or continued web deployment

Contact & Support

Channel	Detail
Email	support@deex.app
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Organization	DeeX Technologies Ltd (DEEX OPTIONS LTD)

This document reflects the state of the DeeX frontend prototype as of April 2, 2026. Metrics and features described herein are based on mock data and UI implementations. Production capabilities depend on the separate backend system.

2. User Stories

DeeX — User Stories

Module: Authentication & Onboarding

US-AUTH-001: View Onboarding Slides

As a new user opening the app for the first time,
I want to see a 3-slide walkthrough of DeeX features,
so that I understand what the app does before signing up.

Priority: High

Acceptance Criteria:

Given I have not completed onboarding

When I open the app

Then I see 3 slides: Trade Crypto, Pay Bills & Send Money, Earn Rewards

And I can navigate with Next/Skip buttons

And a "Get Started" button appears on the last slide

US-AUTH-002: Complete Onboarding

As a new user,
I want to mark onboarding as complete and proceed to login,
so that I don't see onboarding again.

Priority: High

Acceptance Criteria:

Given I am on the last onboarding slide

When I tap "Get Started"

Then `deex\_onboarded` is set in localStorage

And I am navigated to /login

US-AUTH-003: Login with Email and Password

As a registered user,
I want to log in with my email and password,
so that I can access my account.

Priority: High

Acceptance Criteria:

Given I am on the login screen

When I enter my email and password and tap "Log In"

Then I am navigated to the PIN lock screen

US-AUTH-004: Sign Up for New Account

As a new user,
I want to create an account with my name, email, username, and password,
so that I can start using DeeX.

Priority: High

Acceptance Criteria:

Given I am on the signup screen
When I fill in all fields and tap "Create Account"
Then I am navigated to the dashboard

US-AUTH-005: Enter PIN to Unlock

As a logged-in user,
I want to enter my 4-digit PIN to access the app,
so that my account is protected from unauthorized access.

Priority: High

Acceptance Criteria:

Given I am on the PIN lock screen
When I enter the correct 4-digit PIN
Then I am navigated to the dashboard after a brief delay
And if I enter an incorrect PIN, I see an error message and the PIN resets

US-AUTH-006: Social Login Options

As a user who prefers social login,
I want to sign in with Google or Apple,
so that I don't need to remember a password.

Priority: Medium

Acceptance Criteria:

Given I am on the login or signup screen
When I tap the Google or Apple button
Then the respective OAuth flow is initiated (backend integration required)

US-AUTH-007: Biometric Authentication

As a user with a device that supports biometrics,
I want to use fingerprint/Face ID to unlock the app,
so that I can access it faster without typing a PIN.

Priority: Medium

Acceptance Criteria:

Given I am on the PIN lock screen
When I tap the fingerprint icon
Then the device's biometric prompt is shown
And on success, I am navigated to the dashboard

US-AUTH-008: Switch Account from PIN Lock

As a user who wants to log into a different account,
I want to switch accounts from the PIN lock screen,
so that I don't need to navigate back manually.

Priority: Low

Acceptance Criteria:

Given I am on the PIN lock screen

When I tap "Switch Account"

Then I am navigated back to the login screen

US-AUTH-009: Forgot Password

As a user who forgot their password,
I want to initiate a password reset,
so that I can regain access to my account.

Priority: High

Acceptance Criteria:

Given I am on the login screen

When I tap "Forgot Password?"

Then I am taken to a password reset flow (email OTP or link)

US-AUTH-010: Logout

As a user,
I want to log out of my account,
so that my session is securely terminated.

Priority: High

Acceptance Criteria:

Given I am logged in

When I navigate to Profile and tap "Log Out"

Then I am navigated to the login screen

And my session data is cleared

Module: Crypto Trading

US-TRADE-001: View Live Crypto Rates

As a trader,
I want to see a scrolling ticker of live crypto-to-NGN rates,
so that I can make informed trading decisions.

Priority: High

Acceptance Criteria:

Given I am on the dashboard

When I view the "Today's Rates" section

Then I see a horizontally scrolling ticker with BTC, ETH, USDT, USDC, SOL, BNB, TRX

And each rate shows the price in NGN and 24h change percentage

US-TRADE-002: Deposit Crypto

As a user,

I want to get a deposit address for my crypto,

so that I can receive crypto into my DeeX wallet.

Priority: High

Acceptance Criteria:

Given I tap "Deposit" from the dashboard

When I select a cryptocurrency

Then I see a QR code, wallet address, and network selector

And I can copy the address to clipboard

And I see a warning about sending only the correct crypto on the correct network

US-TRADE-003: Sell Crypto from DeeX Wallet

As a user with crypto in my DeeX wallet,

I want to sell it and receive NGN in my bank account,

so that I can cash out my crypto holdings.

Priority: High

Acceptance Criteria:

Given I tap "Sell" and select "DeeX Wallet"

When I select an asset, enter an amount, and choose a destination bank

Then I see a review page with the NGN equivalent

And after confirmation, I see an "Awaiting Deposit" status page

US-TRADE-004: Sell Crypto from External Wallet

As a user without crypto in DeeX,

I want to send crypto from an external wallet and get paid in NGN,

so that I can trade without pre-funding my DeeX wallet.

Priority: High

Acceptance Criteria:

Given I tap "Sell" and select "External Wallet"

When I select an asset, network, and destination bank

Then I see a deposit address and QR code

And I am told to send any amount and will be paid based on what is received

US-TRADE-005: Swap Crypto Assets

As a user,
I want to swap one crypto for another within my wallet,
so that I can rebalance my portfolio without cashing out.

Priority: High

Acceptance Criteria:

Given I navigate to the Swap screen
When I select from/to assets and enter an amount
Then I see the estimated output amount, exchange rate, and 0.5% fee
And after confirmation, I see a success screen

US-TRADE-006: Withdraw Crypto to External Wallet

As a user,
I want to withdraw crypto from my DeeX wallet to an external address,
so that I can move my crypto elsewhere.

Priority: High

Acceptance Criteria:

Given I navigate to the Withdraw screen
When I select an asset, enter a wallet address, network, and amount
Then I see a confirmation page with network fee
And I must enter my PIN to confirm the withdrawal

US-TRADE-007: Send Money to Bank Account

As a user,
I want to send NGN from my crypto wallet to any Nigerian bank account,
so that I can transfer funds easily.

Priority: High

Acceptance Criteria:

Given I navigate to the Send Money screen
When I enter account number, select bank, choose wallet source, and enter amount
Then I see a daily limit tracker
And I cannot exceed the ₦50,000 daily limit

US-TRADE-008: View Asset Details

As a user,
I want to tap on any asset in my wallet to see its details,
so that I can view balance and perform actions on it.

Priority: Medium

Acceptance Criteria:

Given I tap on an asset in the Wallet page
When I am on the Asset Detail page
Then I see my balance, current rate, and action buttons (Deposit, Sell, Swap, Withdraw)

US-TRADE-009: View Wallet Balance with Privacy Toggle

As a user,
I want to toggle visibility of my wallet balance,
so that I can hide it from onlookers in public.

Priority: Medium

Acceptance Criteria:

Given I am on the Wallet or Dashboard page

When I tap the eye icon

Then my balance is masked with "....."

And tapping again reveals the balance

US-TRADE-010: View Transaction Receipt

As a user,
I want to view a receipt for any completed transaction,
so that I have proof of the transaction.

Priority: Medium

Acceptance Criteria:

Given I tap on a transaction from Activity

When I am on the Receipt page

Then I see transaction details including type, amount, date, status, and hash ID

Module: Gift Cards

US-GC-001: Browse Gift Card Brands

As a user who wants to sell a gift card,
I want to browse available gift card brands with their rates,
so that I can choose which card to sell.

Priority: High

Acceptance Criteria:

Given I navigate to the Gift Cards page

When I view the brand grid

Then I see all enabled brands (Apple, Google Play, Amazon, Steam, etc.) with their best rates

US-GC-002: Configure Gift Card Details

As a user selling a gift card,
I want to select the country, card type, and value,
so that the system can calculate my payout.

Priority: High

Acceptance Criteria:

Given I select a brand

When I choose country, card type, and enter a value

Then I see the applicable rate tier and calculated NGN payout

US-GC-003: Upload Gift Card or Enter Code

As a user selling a gift card,

I want to either upload an image of the card or enter the card code,

so that the admin can verify and process my trade.

Priority: High

Acceptance Criteria:

Given I am on the upload step

When I tap the upload area or enter a card code

Then the system records my submission

And I can proceed to review

US-GC-004: Submit Gift Card Trade

As a user,

I want to review and submit my gift card trade,

so that the admin can process it and credit my account.

Priority: High

Acceptance Criteria:

Given I am on the review page

When I confirm the trade details and tap "Submit Trade"

Then the order is saved to localStorage

And I see a "Trade Submitted" status page with estimated processing time

Module: Bill Payments

US-BILL-001: Buy Airtime

As a user,

I want to buy airtime for any Nigerian network,

so that I can top up my phone or someone else's.

Priority: High

Acceptance Criteria:

Given I navigate to Bills → Airtime

When I select a provider (MTN, Glo, Airtel, 9mobile), enter phone number and amount

Then I see a review page and can confirm the purchase

US-BILL-002: Buy Data Plan

As a user,
I want to buy a mobile data plan,
so that I can get internet access.

Priority: High

Acceptance Criteria:

Given I navigate to Bills → Data
When I select a provider, choose a plan, and enter phone number
Then I see a review page and can confirm

US-BILL-003: Pay Electricity Bill

As a user,
I want to pay my electricity bill,
so that I can keep my power supply active.

Priority: High

Acceptance Criteria:

Given I navigate to Bills → Electricity
When I select a Disco (IKEDC, EKEDC, AEDC, PHED, BEDC), enter meter number and amount
Then I see a review page and can confirm

US-BILL-004: Fund Betting Account

As a user,
I want to fund my betting account,
so that I can place bets.

Priority: Medium

Acceptance Criteria:

Given I navigate to Bills → Betting
When I select a provider (Bet9ja, SportyBet, 1xBet, BetKing, MSport), enter user ID and amount
Then I see a review page and can confirm

US-BILL-005: Use Beneficiaries for Quick Payment

As a frequent bill payer,
I want to select from my recent beneficiaries,
so that I don't have to re-enter details each time.

Priority: Medium

Acceptance Criteria:

Given I am on a bill payment form
When I view the beneficiaries section
Then I see recent phone numbers/meter IDs with labels
And tapping one auto-fills the form

Module: DeeX Pay

US-PAY-001: Pay Merchant

As a user,

I want to pay a merchant using their ID, QR code, or payment link,
so that I can make in-store or online payments.

Priority: High

Acceptance Criteria:

Given I navigate to DeeX Pay → Pay Merchant

When I enter a merchant ID, scan a QR, or paste a link

Then I can select which crypto to pay with and enter the amount

US-PAY-002: Send Cash to Bank

As a user,

I want to convert my crypto and send NGN to any bank account,
so that I can pay someone who doesn't use crypto.

Priority: High

Acceptance Criteria:

Given I navigate to DeeX Pay → Send Cash

When I select a wallet source, bank, account number, and amount

Then I see my daily limit tracker (₦100k/day)

And I cannot exceed the remaining limit

US-PAY-003: Generate Payment Link

As a user or merchant,

I want to generate a shareable payment link,
so that I can receive crypto payments from anyone.

Priority: Medium

Acceptance Criteria:

Given I navigate to DeeX Pay → Generate Link

When I enter an amount in USD and optional description

Then a unique payment link is generated

And I can copy and share it

Module: Virtual Cards

US-VC-001: Create Virtual Card

As a user with KYC Level 2+,

I want to create a Visa virtual card,
so that I can make online payments internationally.

Priority: High

Acceptance Criteria:

Given I navigate to Virtual Cards

When I tap "Create New Card"

Then I enter a label and the \$2 creation fee is deducted from my selected wallet

And a new card is created with a random 4-digit last number

US-VC-002: Fund Virtual Card

As a cardholder,

I want to add funds to my virtual card from my crypto wallet,

so that I can use it for purchases.

Priority: High

Acceptance Criteria:

Given I select a card and tap "Fund"

When I enter an amount in USD and select a source wallet

Then the card balance is updated

US-VC-003: Freeze/Unfreeze Card

As a cardholder,

I want to freeze my card temporarily,

so that no unauthorized transactions can occur.

Priority: High

Acceptance Criteria:

Given I view a card

When I tap "Freeze"

Then the card status changes to "Frozen" and an overlay appears on the card visual

And tapping "Unfreeze" restores it to active

US-VC-004: View Card Details

As a cardholder,

I want to view my full card number, expiry, and CVV,

so that I can use the card for online payments.

Priority: High

Acceptance Criteria:

Given I view a card

When I tap the eye icon

Then the full card number, expiry date, and CVV are revealed

And I can copy the card number

US-VC-005: Manage Spending Limits

As a cardholder,
I want to set daily and monthly spending limits on my card,
so that I can control my spending.

Priority: Medium

Acceptance Criteria:

Given I view a card and tap "Limits"

When I enter new daily and monthly limits

Then the limits are saved and displayed on the card detail page

Module: Rewards & Referrals

US-REW-001: View DeeXPoints Balance

As a user,
I want to see my DeeXPoints balance and its NGN value,
so that I know how much I've earned.

Priority: High

Acceptance Criteria:

Given I navigate to Rewards

When I view the DeeXPoints card

Then I see my points balance and the equivalent NGN value (1pt = ₦10)

US-REW-002: Redeem Points for Cash

As a user with sufficient points,
I want to redeem my points for NGN cash,
so that I can withdraw my rewards.

Priority: High

Acceptance Criteria:

Given I have at least 500 points

When I navigate to Redeem and enter an amount

Then I see the cash value and confirm the redemption

And the redemption is submitted for admin approval

US-REW-003: View Referral Dashboard

As a user,
I want to see my referral stats, link, and list of referred users,
so that I can track my referral earnings.

Priority: High

Acceptance Criteria:

Given I navigate to Referrals

When I view the dashboard

Then I see total referrals, earned amount, pending amount, my referral link, and a list of referred users

US-REW-004: View Leaderboard

As a competitive user,

I want to see the referral leaderboard,

so that I can see how I rank against others.

Priority: Low

Acceptance Criteria:

Given I view the Referrals page

When I scroll to the Leaderboard section

Then I see the top referrers ranked by number of referrals

US-REW-005: Track Trade Streak

As a daily trader,

I want to see my trade streak and milestone progress,

so that I can maximize my streak bonus points.

Priority: Medium

Acceptance Criteria:

Given I navigate to Trade Streak

When I view the overview

Then I see my current streak, longest streak, multiplier, and points earned

And I see a weekly calendar with fire icons for completed days

US-REW-006: View Earnings Log

As a user,

I want to see a detailed log of how I earned each point,

so that I can track my reward sources.

Priority: Medium

Acceptance Criteria:

Given I navigate to Rewards → Earnings Log

When I view the list

Then I see each earning entry with source, category, points, and date

Module: KYC Verification

US-KYC-001: View KYC Overview

As a user,

I want to see my current KYC status and available levels,
so that I know what I need to do to increase my limits.

Priority: High

Acceptance Criteria:

Given I navigate to KYC Verification

When I view the overview

Then I see my current trading/withdrawal limits, progress bar, and all 3 KYC levels with their requirements

US-KYC-002: Submit KYC Level 1

As a new user,

I want to submit my BVN and complete a liveness check,
so that I can unlock basic trading (\$1,000 limit).

Priority: High

Acceptance Criteria:

Given I tap on KYC Level 1

When I enter my 11-digit BVN and complete the selfie liveness check

Then my submission is sent for review

US-KYC-003: Submit KYC Level 2

As a user with Level 1,

I want to upload my government ID and proof of address,
so that I can increase my limits to \$10,000 trading.

Priority: High

Acceptance Criteria:

Given I tap on KYC Level 2

When I upload a government ID and proof of address document

Then my submission is sent for review

US-KYC-004: Submit KYC Level 3

As a user with Level 2,

I want to submit employment docs, source of income, and risk questionnaire,
so that I can unlock the highest limits (\$25,000 trading).

Priority: High

Acceptance Criteria:

Given I tap on KYC Level 3

When I upload employment docs, select income source, complete the risk questionnaire, and have 2FA enabled

Then my submission is sent for review

Module: Notifications

US-NOTIF-001: View Notifications

As a user,

I want to see all my notifications in one place,

so that I stay informed about my account.

Priority: High

Acceptance Criteria:

Given I tap the notification bell

When I view the notifications page

Then I see login alerts, session terminations, and promotional messages with timestamps

US-NOTIF-002: Receive Login Alerts

As a security-conscious user,

I want to be notified whenever I log in,

so that I can detect unauthorized access.

Priority: High

Acceptance Criteria:

Given I log into my account

When the login is successful

Then a login notification is created with device and time info

US-NOTIF-003: Receive Session Termination Alerts

As a user,

I want to be notified when my previous session is terminated,

so that I know if someone else logged into my account.

Priority: High

Acceptance Criteria:

Given I log in from a new device

When my previous session is terminated

Then I receive a session termination notification

Module: Settings & Security

US-SET-001: Change PIN

As a user,

I want to change my 4-digit PIN,

so that I can maintain account security.

Priority: High

Acceptance Criteria:

Given I navigate to Security settings

When I tap "Change PIN"
Then I am prompted to enter my current PIN and a new PIN

US-SET-002: Toggle Biometric Login

As a user,
I want to enable or disable biometric login,
so that I can choose how I authenticate.

Priority: Medium

Acceptance Criteria:

Given I am in Security settings
When I toggle "Biometric Login"
Then the setting is saved and reflected in the UI

US-SET-003: Toggle Two-Factor Authentication

As a security-conscious user,
I want to enable 2FA on my account,
so that my account has an extra layer of protection.

Priority: High

Acceptance Criteria:

Given I am in Security settings
When I toggle "Two-Factor Auth"
Then I am guided through the 2FA setup process

US-SET-004: View Active Sessions

As a user,
I want to see all my active sessions,
so that I can detect and terminate unauthorized access.

Priority: High

Acceptance Criteria:

Given I view the Active Sessions section
When I see my sessions listed
Then each shows device, location, and time
And I can terminate non-current sessions

US-SET-005: View Login History

As a user,
I want to see my recent login attempts,
so that I can spot any suspicious activity.

Priority: Medium

Acceptance Criteria:

Given I view the Login History section

When I see the list

Then each entry shows device, date, time, and success/failure status

US-SET-006: Manage Bank Accounts

As a user,

I want to manage my linked bank accounts,

so that I can receive payouts to the right account.

Priority: High

Acceptance Criteria:

Given I navigate to Manage Beneficiaries

When I view my linked accounts

Then I can add, edit, or remove bank accounts

US-SET-007: Generate Account Statement

As a user,

I want to generate a statement of my transactions,

so that I can use it for record-keeping or tax purposes.

Priority: Medium

Acceptance Criteria:

Given I navigate to Generate Statement

When I select a date range

Then a statement is generated with all my transactions

US-SET-008: Access Help & Support

As a user who needs help,

I want to contact support through multiple channels,

so that I can get my issues resolved.

Priority: High

Acceptance Criteria:

Given I navigate to Help & Support

When I view the contact options

Then I see Live Chat, Email (support@deex.app), and WhatsApp (+234 810 367 4006)

And I can also browse FAQs

Module: Admin Dashboard

US-ADM-001: View Admin Dashboard Overview

As an admin,
I want to see key platform metrics at a glance,
so that I can monitor platform health.

Priority: High

Acceptance Criteria:

Given I navigate to the Admin Dashboard

When I view the overview

Then I see total payout, active users, total customers, merchants, and performance charts

US-ADM-002: Manage User Accounts

As an admin,
I want to view and search all users,
so that I can manage customer accounts.

Priority: High

Acceptance Criteria:

Given I navigate to Admin → Users

When I view the customer list

Then I can search by name/email, filter by status, and click into user details

US-ADM-003: Approve/Reject KYC

As an admin,
I want to review and approve or reject KYC submissions,
so that I can verify user identities and set appropriate limits.

Priority: High

Acceptance Criteria:

Given I navigate to Admin → KYC

When I view pending submissions

Then I can approve (updating limits) or reject (with reason) each submission

US-ADM-004: Monitor Compliance Alerts

As an admin,
I want to view and act on compliance alerts,
so that I can prevent fraud and maintain regulatory compliance.

Priority: High

Acceptance Criteria:

Given I navigate to Admin → Compliance

When I view alerts

Then I can filter by severity/status, view detailed timelines, approve/dismiss, or suspend users

US-ADM-005: Configure Compliance Rules

As an admin,
I want to configure auto-suspension rules,
so that the system can automatically flag suspicious activity.

Priority: High

Acceptance Criteria:

Given I navigate to Compliance → Rules Engine

When I view the rules list

Then I can toggle rules on/off and see thresholds, last triggered date, and trigger counts

US-ADM-006: Manage Gift Card Orders

As an admin,
I want to approve or reject gift card trade orders,
so that I can verify card validity before crediting users.

Priority: High

Acceptance Criteria:

Given I navigate to Admin → Gift Cards

When I view pending orders

Then I can view card images/codes, approve (crediting the user), or reject with notes

US-ADM-007: Manage Gift Card Brands & Rates

As an admin,
I want to configure gift card brands and their rate tiers,
so that I can control which cards are accepted and at what rates.

Priority: High

Acceptance Criteria:

Given I navigate to Gift Cards → Brands & Rates

When I view the brand list

Then I can enable/disable brands, edit rate tiers per country, and add new brands

US-ADM-008: Monitor Wallets

As an admin,
I want to view DeeX and customer wallet balances and activity,
so that I can monitor fund flows.

Priority: High

Acceptance Criteria:

Given I navigate to Admin → Wallets

When I toggle between DeeX and Customer wallets

Then I see total crypto, asset breakdowns, trade volume charts, and activity logs

US-ADM-009: View Audit Log

As an admin,
I want to see a log of all admin actions,
so that I can maintain accountability and compliance.

Priority: High

Acceptance Criteria:

Given I navigate to Admin → Audit Log

When I view the log

Then I see each action with admin name, action type, target, details, and timestamp

US-ADM-010: Manage Rewards Configuration

As an admin,
I want to configure reward rules and approve redemptions,
so that I can control the rewards program.

Priority: Medium

Acceptance Criteria:

Given I navigate to Admin → Rewards

When I view the config tab

Then I can see and edit reward triggers and amounts

And in the payouts tab, I can approve or reject redemption requests

US-ADM-011: Manage Influencers

As an admin,
I want to appoint and manage influencers with custom profit-share,
so that I can run a special referral program.

Priority: Medium

Acceptance Criteria:

Given I navigate to Rewards → Influencers

When I view the influencer list

Then I can appoint new influencers, edit their profit-share %, and pause/remove them

US-ADM-012: View Reports

As an admin,
I want to view platform reports,
so that I can analyze performance and make data-driven decisions.

Priority: Medium

Acceptance Criteria:

Given I navigate to Admin → Reports

When I select a report type

Then I see charts and metrics for Revenue, Growth, Retention, Website, Happiness, or Survey

US-ADM-013: Configure Platform Fees

As an admin,
I want to configure platform fees,
so that I can control DeeX's revenue.

Priority: High

Acceptance Criteria:

Given I navigate to Settings → Fees

When I view the fee list

Then I can see and edit Crypto Trade Fee, Gift Card Fee, Withdrawal Fee, and DeeX Pay Fee

US-ADM-014: Manage Payroll

As an admin,
I want to manage staff payroll,
so that I can track employee compensation.

Priority: Low

Acceptance Criteria:

Given I navigate to Settings → Payroll

When I view the staff list

Then I can see each staff member's name, email, role, salary, and date created

And I can add or remove staff

3. QA Test Cases

DeeX — QA Test Cases

Smoke Test Suite (Top 20 Critical Tests)

#	Test Case	Module	Priority	Status
1	TC-AUTH-001: Onboarding completes and navigates to login	Auth	High	Pending
2	TC-AUTH-003: Login navigates to PIN lock	Auth	High	Pending
3	TC-AUTH-005: Correct PIN unlocks to dashboard	Auth	High	Pending
4	TC-AUTH-005: Incorrect PIN shows error	Auth	High	Pending
5	TC-TRADE-001: Dashboard shows crypto rates ticker	Trading	High	Pending
6	TC-TRADE-002: Deposit shows QR and address	Trading	High	Pending
7	TC-TRADE-003: Sell from wallet flow completes	Trading	High	Pending
8	TC-TRADE-005: Swap calculates output correctly	Trading	High	Pending
9	TC-GC-001: Gift card brands display with rates	Gift Cards	High	Pending
10	TC-GC-004: Gift card trade submits successfully	Gift Cards	High	Pending
11	TC-BILL-001: Airtime purchase flow completes	Bill Pay	High	Pending
12	TC-PAY-001: Pay merchant flow navigates correctly	DeeX Pay	High	Pending
13	TC-VC-001: Virtual card creation works	Virtual Cards	High	Pending
14	TC-REW-001: Rewards page shows points balance	Rewards	High	Pending
15	TC-KYC-001: KYC overview displays all levels	KYC	High	Pending
16	TC-ADM-001: Admin dashboard loads with metrics	Admin	High	Pending
17	TC-ADM-003: KYC approve/reject works	Admin	High	Pending
18	TC-ADM-006: Gift card order approval works	Admin	High	Pending
19	TC-SEC-001: PIN brute force does not allow access	Security	High	Pending
20	TC-NAV-001: Bottom navigation switches pages	Navigation	High	Pending

Authentication Test Cases

TC-AUTH-001: Onboarding Flow Completes

Field	Value
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Module	Authentication
Priority	High
Type	Functional
Preconditions	`deex_onboarded` not set in localStorage
Test Steps	1. Open app at root URL 2. Verify redirected to /onboarding 3. Verify 3 slides display 4. Tap "Next" twice 5. Tap "Get Started"
Expected Result	`deex_onboarded` = "1" in localStorage, navigated to /login
Actual Result	_[Leave blank]_
Status	 Pending

TC-AUTH-002: Skip Onboarding

Field	Value
Module	Authentication
Priority	Medium
Type	Functional
Preconditions	On first slide of onboarding
Test Steps	1. Tap "Skip" button 2. Verify navigation
Expected Result	`deex_onboarded` = "1" in localStorage, navigated to /login
Actual Result	_[Leave blank]_
Status	 Pending

TC-AUTH-003: Login Navigation

Field	Value
Module	Authentication
Priority	High
Type	Functional
Preconditions	Onboarding completed
Test Steps	1. Navigate to /login 2. Enter any email 3. Enter any password 4. Tap "Log In"

Expected Result	Navigated to /pin (PIN lock screen)
Actual Result	_[Leave blank]_
Status	Pending

TC-AUTH-004: Signup Navigation

Field	Value
Module	Authentication
Priority	High
Type	Functional
Preconditions	On /login page
Test Steps	1. Tap "Sign Up" 2. Verify /signup loads 3. Tap "Create Account"
Expected Result	Navigated to /dashboard
Actual Result	_[Leave blank]_
Status	Pending

TC-AUTH-005: Correct PIN Entry

Field	Value
Module	Authentication
Priority	High
Type	Functional
Preconditions	On /pin screen
Test Steps	1. Enter "1234" on PIN keypad 2. Wait 300ms
Expected Result	Navigated to /dashboard
Actual Result	_[Leave blank]_
Status	Pending

TC-AUTH-006: Incorrect PIN Entry

Field	Value
Module	Authentication

Priority	High
Type	Functional
Preconditions	On /pin screen
Test Steps	1. Enter "0000" on PIN keypad 2. Wait 300ms
Expected Result	Error message "Incorrect PIN. Try again." displayed, PIN dots reset
Actual Result	_[Leave blank]_
Status	 Pending

TC-AUTH-007: PIN Delete Button

Field	Value
Module	Authentication
Priority	Medium
Type	Functional
Preconditions	On /pin screen, entered 2 digits
Test Steps	1. Tap delete button
Expected Result	Last digit removed, 1 dot filled
Actual Result	_[Leave blank]_
Status	 Pending

TC-AUTH-008: Switch Account from PIN

Field	Value
Module	Authentication
Priority	Low
Type	Functional
Preconditions	On /pin screen
Test Steps	1. Tap "Switch Account"
Expected Result	Navigated to /login
Actual Result	_[Leave blank]_
Status	 Pending

TC-AUTH-009: Social Login Buttons Render

Field	Value
Module	Authentication
Priority	Medium
Type	UI
Preconditions	On /login or /signup
Test Steps	1. Verify Google button renders 2. Verify Apple button renders 3. Tap Google button
Expected Result	Buttons render correctly; tap does nothing (no backend integration yet)
Actual Result	_[Leave blank]_
Status	 Pending

TC-AUTH-010: Logout Flow

Field	Value
Module	Authentication
Priority	High
Type	Functional
Preconditions	Logged in, on any page
Test Steps	1. Navigate to /profile 2. Tap "Log Out"
Expected Result	Navigated to /login
Actual Result	_[Leave blank]_
Status	 Pending

TC-AUTH-011: Forgot Password Link

Field	Value
Module	Authentication
Priority	Medium
Type	Functional
Preconditions	On /login
Test Steps	1. Tap "Forgot Password?"
Expected Result	Navigation triggered (page may not exist yet — expected gap)
Actual Result	_[Leave blank]_

Status	Pending
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TC-AUTH-012: Biometric Button Renders

Field	Value
Module	Authentication
Priority	Low
Type	UI
Preconditions	On /pin screen
Test Steps	1. Verify fingerprint icon renders in keypad
Expected Result	Fingerprint icon visible in bottom-left of keypad grid
Actual Result	_[Leave blank]_
Status	Pending

TC-AUTH-013: Onboarding Already Completed

Field	Value
Module	Authentication
Priority	Medium
Type	Functional
Preconditions	`deex_onboarded` = "1" in localStorage
Test Steps	1. Open app at root URL
Expected Result	Redirected to /login (not /onboarding)
Actual Result	_[Leave blank]_
Status	Pending

TC-AUTH-014: PIN Keypad Overflow Protection

Field	Value
Module	Authentication
Priority	Medium
Type	Edge
Preconditions	On /pin screen, already entered 4 digits
Test Steps	1. Tap additional digit

Expected Result	No additional digits accepted (pin.length >= 4 guard)
Actual Result	_[Leave blank]_
Status	Pending

TC-AUTH-015: Default PIN Hint Display

Field	Value
Module	Authentication
Priority	Low
Type	UI
Preconditions	On /pin screen, no error
Test Steps	1. Verify "Default PIN: 1234" text is visible
Expected Result	Hint text visible below PIN dots
Actual Result	_[Leave blank]_
Status	Pending

Crypto Trading Test Cases

TC-TRADE-001: Dashboard Crypto Rates Ticker

Field	Value
Module	Crypto Trading
Priority	High
Type	UI
Preconditions	On /dashboard
Test Steps	1. Scroll to "Today's Rates" section 2. Verify ticker auto-scrolls
Expected Result	Ticker shows BTC, ETH, USDT, USDC, SOL, BNB, TRX with rates and change %
Actual Result	_[Leave blank]_
Status	Pending

TC-TRADE-002: Deposit Crypto Selection

Field	Value
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Module	Crypto Trading
Priority	High
Type	Functional
Preconditions	On /dashboard, tap "Deposit"
Test Steps	1. Verify /deposit loads 2. Verify BTC, ETH, USDT, USDC, SOL listed 3. Tap USDT
Expected Result	Navigated to address view for USDT
Actual Result	_[Leave blank]_
Status	 Pending

TC-TRADE-003: Deposit Address Display

Field	Value
Module	Crypto Trading
Priority	High
Type	Functional
Preconditions	On deposit address view for selected crypto
Test Steps	1. Verify QR code placeholder displays 2. Verify wallet address displays 3. Verify network selector shows
Expected Result	QR code, address, and network selector all visible
Actual Result	_[Leave blank]_
Status	 Pending

TC-TRADE-004: Deposit Copy Address

Field	Value
Module	Crypto Trading
Priority	High
Type	Functional
Preconditions	On deposit address view
Test Steps	1. Tap copy button 2. Wait 2 seconds
Expected Result	Address copied to clipboard, checkmark icon shown temporarily
Actual Result	_[Leave blank]_

Status	 Pending
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TC-TRADE-005: Deposit Network Switch

Field	Value
Module	Crypto Trading
Priority	High
Type	Functional
Preconditions	On deposit address view for USDT
Test Steps	1. Tap network dropdown 2. Select TRC-20
Expected Result	Network selector updates to TRC-20, dropdown closes
Actual Result	_[Leave blank]_
Status	 Pending

TC-TRADE-006: Deposit Warning Display

Field	Value
Module	Crypto Trading
Priority	Medium
Type	UI
Preconditions	On deposit address view
Test Steps	1. Scroll to warning section
Expected Result	Warning about sending only correct crypto on correct network is visible
Actual Result	_[Leave blank]_
Status	 Pending

TC-TRADE-007: Sell from Wallet Flow

Field	Value
Module	Crypto Trading
Priority	High
Type	Functional
Preconditions	On /dashboard, tap "Sell"

Test Steps	1. Select "DeeX Wallet" 2. Select asset (USDT) 3. Select bank (PalmPay) 4. Select network (BEP-20) 5. Enter amount "100" 6. Tap "Proceed"
Expected Result	NGN calculation shows ₦153,500, navigated to review page
Actual Result	_[Leave blank]_
Status	 Pending

TC-TRADE-008: Sell from External Wallet

Field	Value
Module	Crypto Trading
Priority	High
Type	Functional
Preconditions	On /sell-crypto
Test Steps	1. Select "External Wallet" 2. Select asset, bank, network 3. Verify deposit address shown
Expected Result	Deposit address, QR code, and "I've Sent the Crypto" button shown
Actual Result	_[Leave blank]_
Status	 Pending

TC-TRADE-009: Sell Amount Calculation

Field	Value
Module	Crypto Trading
Priority	High
Type	Functional
Preconditions	On sell form, USDT selected
Test Steps	1. Enter "50" in amount field
Expected Result	NGN equivalent shows ₦76,750 (50 × 1,535)
Actual Result	_[Leave blank]_
Status	 Pending

TC-TRADE-010: Sell Review Page

Field	Value
Module	Crypto Trading
Priority	High
Type	Functional
Preconditions	On sell form, entered amount
Test Steps	1. Tap "Proceed"
Expected Result	Review page shows asset, network, amount, destination, rate, and NGN amount
Actual Result	_[Leave blank]_
Status	 Pending

TC-TRADE-011: Sell Pending State

Field	Value
Module	Crypto Trading
Priority	High
Type	Functional
Preconditions	On sell review, tap "Proceed to Deposit" then "I've Sent the Crypto"
Test Steps	1. Verify pending state page
Expected Result	"Awaiting Deposit" page with Clock icon and "Back to Home" button
Actual Result	_[Leave blank]_
Status	 Pending

TC-TRADE-012: Swap Asset Selection

Field	Value
Module	Crypto Trading
Priority	High
Type	Functional
Preconditions	On /swap-crypto
Test Steps	1. Tap "From" asset selector 2. Select ETH 3. Tap "To" asset selector 4. Select USDT
Expected Result	From shows ETH, To shows USDT, swap button visible between them

Actual Result	_[Leave blank]_
Status	 Pending

TC-TRADE-013: Swap Amount Calculation

Field	Value
Module	Crypto Trading
Priority	High
Type	Functional
Preconditions	On /swap-crypto, From=BTC (rate 97450), To=USDT (rate 1)
Test Steps	1. Enter "0.01" in From amount
Expected Result	To amount shows 974.50 USDT, fee shows 0.000050 BTC (0.5%)
Actual Result	_[Leave blank]_
Status	 Pending

TC-TRADE-014: Swap Confirmation

Field	Value
Module	Crypto Trading
Priority	High
Type	Functional
Preconditions	On /swap-crypto, entered amount > 0
Test Steps	1. Tap "Preview Swap" 2. Verify confirmation page
Expected Result	Shows From, To, Rate, Fee (0.5%), You Receive, and "Swap Now" button
Actual Result	_[Leave blank]_
Status	 Pending

TC-TRADE-015: Swap Success

Field	Value
Module	Crypto Trading
Priority	High
Type	Functional

Preconditions	On swap confirmation page
Test Steps	1. Tap "Swap Now"
Expected Result	Success page with "Swap Successful!" and "Back to Wallet" button
Actual Result	_[Leave blank]_
Status	 Pending

TC-TRADE-016: Withdraw Address Input

Field	Value
Module	Crypto Trading
Priority	High
Type	Functional
Preconditions	On /withdraw, selected BTC
Test Steps	1. Enter wallet address 2. Select network 3. Enter amount
Expected Result	All fields populated, "Preview Withdrawal" button enabled
Actual Result	_[Leave blank]_
Status	 Pending

TC-TRADE-017: Withdraw PIN Confirmation

Field	Value
Module	Crypto Trading
Priority	High
Type	Functional
Preconditions	On withdraw confirmation, tap "Confirm Withdrawal"
Test Steps	1. Enter correct PIN "1234"
Expected Result	Withdrawal submitted, success page shown
Actual Result	_[Leave blank]_
Status	 Pending

TC-TRADE-018: Send Money Daily Limit

Field	Value
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Module	Crypto Trading
Priority	High
Type	Functional
Preconditions	On /send-money
Test Steps	1. Verify daily limit banner shows ₦50,000 2. Enter amount > 50000
Expected Result	Daily limit tracker visible; no explicit validation on amount input (gap)
Actual Result	_[Leave blank]_
Status	 Pending

TC-TRADE-019: Asset Detail Navigation

Field	Value
Module	Crypto Trading
Priority	Medium
Type	Functional
Preconditions	On /wallet
Test Steps	1. Tap on BTC asset
Expected Result	Navigated to /asset/btc
Actual Result	_[Leave blank]_
Status	 Pending

TC-TRADE-020: Balance Privacy Toggle

Field	Value
Module	Crypto Trading
Priority	Medium
Type	UI
Preconditions	On /wallet or /dashboard
Test Steps	1. Tap eye icon 2. Verify balance masked 3. Tap again
Expected Result	Balance toggles between visible and "....."
Actual Result	_[Leave blank]_
Status	 Pending

Gift Cards Test Cases

TC-GC-001: Brand Grid Display

Field	Value
Module	Gift Cards
Priority	High
Type	UI
Preconditions	On /giftcards
Test Steps	1. Verify brand grid displays
Expected Result	Apple, Google Play, Amazon, Steam, iTunes, Walmart, Nike, Sephora shown with rates
Actual Result	_[Leave blank]_
Status	Pending

TC-GC-002: Country Selection

Field	Value
Module	Gift Cards
Priority	High
Type	Functional
Preconditions	Selected Apple brand
Test Steps	1. Tap USA, UK, Canada, EU country pills
Expected Result	Selected country highlighted, rate tiers update
Actual Result	_[Leave blank]_
Status	Pending

TC-GC-003: Card Type Selection

Field	Value
Module	Gift Cards
Priority	High
Type	Functional
Preconditions	On card type step

Test Steps	1. Toggle between "Physical Card" and "E-Code"
Expected Result	Selected type highlighted
Actual Result	_[Leave blank]_
Status	Pending

TC-GC-004: Value Input with Quick Denominations

Field	Value
Module	Gift Cards
Priority	High
Type	Functional
Preconditions	On value step
Test Steps	1. Tap "\$100" quick button 2. Verify rate and payout calculated
Expected Result	Value set to 100, rate and NGN payout displayed correctly
Actual Result	_[Leave blank]_
Status	Pending

TC-GC-005: Card Image Upload

Field	Value
Module	Gift Cards
Priority	High
Type	Functional
Preconditions	On upload step
Test Steps	1. Tap upload area
Expected Result	"Image uploaded" confirmation shown
Actual Result	_[Leave blank]_
Status	Pending

TC-GC-006: Card Code Input

Field	Value
Module	Gift Cards

Priority	High
Type	Functional
Preconditions	On upload step
Test Steps	1. Enter card code in text field
Expected Result	Code stored, can proceed to review
Actual Result	_[Leave blank]_
Status	 Pending

TC-GC-007: Review Accuracy

Field	Value
Module	Gift Cards
Priority	High
Type	Functional
Preconditions	On review step
Test Steps	1. Verify all details match input
Expected Result	Brand, country, type, value, rate, and payout all correct
Actual Result	_[Leave blank]_
Status	 Pending

TC-GC-008: Trade Submission to localStorage

Field	Value
Module	Gift Cards
Priority	High
Type	Integration
Preconditions	On review step, tap "Submit Trade"
Test Steps	1. Tap "Submit Trade" 2. Check localStorage for deex_gc_orders
Expected Result	Order saved to localStorage, status page shown
Actual Result	_[Leave blank]_
Status	 Pending

TC-GC-009: Status Page Display

Field	Value
Module	Gift Cards
Priority	High
Type	UI
Preconditions	After submitting trade
Test Steps	1. Verify status page
Expected Result	"Trade Submitted!" with Clock icon, trade summary, "Notified Admin" badge
Actual Result	_[Leave blank]_
Status	 Pending

TC-GC-010: Receipt Navigation

Field	Value
Module	Gift Cards
Priority	Medium
Type	Functional
Preconditions	On status page
Test Steps	1. Tap "View Receipt"
Expected Result	Navigated to /receipt with giftcard data
Actual Result	_[Leave blank]_
Status	 Pending

TC-GC-011: Back Navigation Between Steps

Field	Value
Module	Gift Cards
Priority	Medium
Type	Functional
Preconditions	On any step after brand selection
Test Steps	1. Tap back button
Expected Result	Returns to previous step in flow
Actual Result	_[Leave blank]_
Status	 Pending

TC-GC-012: Rate Tier Display

Field	Value
Module	Gift Cards
Priority	Medium
Type	UI
Preconditions	On card type step
Test Steps	1. Scroll to rate tiers section
Expected Result	Rate tiers for selected country displayed (min-max → rate)
Actual Result	_[Leave blank]_
Status	 Pending

Bill Payments Test Cases

TC-BILL-001: Airtime Purchase Flow

Field	Value
Module	Bill Payments
Priority	High
Type	Functional
Preconditions	Navigate to /bills/airtime
Test Steps	1. Select MTN provider 2. Enter phone number 3. Enter amount 4. Tap "Continue" 5. Review and "Pay Now"
Expected Result	Success page with CheckCircle icon
Actual Result	_[Leave blank]_
Status	 Pending

TC-BILL-002: Data Plan Selection

Field	Value
Module	Bill Payments

Priority	High
Type	Functional
Preconditions	Navigate to /bills/data
Test Steps	1. Select provider 2. Select data plan (e.g., "2GB - ₦1,000") 3. Enter phone number
Expected Result	Plan selected and highlighted, form complete
Actual Result	_[Leave blank]_
Status	 Pending

TC-BILL-003: Electricity Meter Input

Field	Value
Module	Bill Payments
Priority	High
Type	Functional
Preconditions	Navigate to /bills/electricity
Test Steps	1. Select IKEDC 2. Enter meter number 3. Enter amount
Expected Result	Form fields populated, can proceed to review
Actual Result	_[Leave blank]_
Status	 Pending

TC-BILL-004: Betting Account Funding

Field	Value
Module	Bill Payments
Priority	Medium
Type	Functional
Preconditions	Navigate to /bills/betting
Test Steps	1. Select Bet9ja 2. Enter user ID 3. Enter amount
Expected Result	Form complete, can proceed
Actual Result	_[Leave blank]_

Status	 Pending
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TC-BILL-005: Beneficiary Quick Select

Field	Value
Module	Bill Payments
Priority	Medium
Type	Functional
Preconditions	On airtime form
Test Steps	1. Tap a beneficiary chip
Expected Result	Phone number auto-filled
Actual Result	_[Leave blank]_
Status	 Pending

TC-BILL-006: Provider Grid Display

Field	Value
Module	Bill Payments
Priority	High
Type	UI
Preconditions	On any bill type page, provider step
Test Steps	1. Verify provider grid
Expected Result	2-column grid of providers with icons
Actual Result	_[Leave blank]_
Status	 Pending

TC-BILL-007: Form Validation

Field	Value
Module	Bill Payments
Priority	High
Type	Functional
Preconditions	On bill form with empty fields
Test Steps	1. Tap "Continue" without filling required fields

Expected Result	No explicit validation (gap — should validate)
Actual Result	_[Leave blank]_
Status	 Pending

TC-BILL-008: Review Page Accuracy

Field	Value
Module	Bill Payments
Priority	High
Type	Functional
Preconditions	On review step
Test Steps	1. Verify all fields match input
Expected Result	Service, provider, plan, and form data all correct
Actual Result	_[Leave blank]_
Status	 Pending

TC-BILL-009: Success Page

Field	Value
Module	Bill Payments
Priority	High
Type	UI
Preconditions	After confirming payment
Test Steps	1. Verify success page
Expected Result	CheckCircle icon, "Payment Successful!", service name, amount, "View Receipt" and "Back to Home" buttons
Actual Result	_[Leave blank]_
Status	 Pending

TC-BILL-010: Receipt Navigation

Field	Value
Module	Bill Payments
Priority	Medium

Type	Functional
Preconditions	On success page
Test Steps	1. Tap "View Receipt"
Expected Result	Navigated to /receipt with bill payment data
Actual Result	_[Leave blank]_
Status	Pending

DeeX Pay Test Cases

TC-PAY-001: Pay Merchant Flow

Field	Value
Module	DeeX Pay
Priority	High
Type	Functional
Preconditions	On /deex-pay
Test Steps	1. Tap "Pay Merchant" 2. Enter merchant ID 3. Tap "Continue"
Expected Result	Crypto selection page shown
Actual Result	_[Leave blank]_
Status	Pending

TC-PAY-002: QR Scan Button

Field	Value
Module	DeeX Pay
Priority	Medium
Type	UI
Preconditions	On pay merchant step
Test Steps	1. Verify "Scan QR Code" button visible
Expected Result	Button renders (non-functional without camera integration)
Actual Result	_[Leave blank]_
Status	Pending

TC-PAY-003: Send Cash Form

Field	Value
Module	DeeX Pay
Priority	High
Type	Functional
Preconditions	On /deex-pay, tap "Send Cash to Bank"
Test Steps	1. Select wallet source 2. Select bank 3. Enter account number 4. Enter amount
Expected Result	All fields populated, daily limit tracker visible
Actual Result	_[Leave blank]_
Status	Pending

TC-PAY-004: Daily Limit Display

Field	Value
Module	DeeX Pay
Priority	High
Type	UI
Preconditions	On send cash form
Test Steps	1. Verify daily limit banner
Expected Result	Shows "Daily limit: ₦100,000" with progress bar and "₦35,000 used"
Actual Result	_[Leave blank]_
Status	Pending

TC-PAY-005: Payment Link Generation

Field	Value
Module	DeeX Pay
Priority	Medium
Type	Functional
Preconditions	On /deex-pay, tap "Generate Payment Link"

Test Steps	1. Enter amount 2. Enter description (optional) 3. Tap "Generate Link"
Expected Result	Generated link displayed with copy button
Actual Result	_[Leave blank]_
Status	 Pending

TC-PAY-006: Link Copy

Field	Value
Module	DeeX Pay
Priority	Medium
Type	Functional
Preconditions	Link generated
Test Steps	1. Tap copy button
Expected Result	Link copied, checkmark shown
Actual Result	_[Leave blank]_
Status	 Pending

TC-PAY-007: Exceeds Daily Limit Warning

Field	Value
Module	DeeX Pay
Priority	High
Type	Functional
Preconditions	On send cash form, remaining limit ₦65,000
Test Steps	1. Enter amount > 65000
Expected Result	Warning "Exceeds remaining daily limit" shown
Actual Result	_[Leave blank]_
Status	 Pending

TC-PAY-008: Merchant Payment Success

Field	Value
Module	DeeX Pay

Priority	High
Type	Functional
Preconditions	On merchant payment review
Test Steps	1. Tap "Pay Now"
Expected Result	Success page with "Payment Sent!"
Actual Result	_[Leave blank]_
Status	 Pending

TC-PAY-009: Send Cash Success

Field	Value
Module	DeeX Pay
Priority	High
Type	Functional
Preconditions	On send cash review
Test Steps	1. Tap "Send Now"
Expected Result	Success page with "Transfer Sent!"
Actual Result	_[Leave blank]_
Status	 Pending

TC-PAY-010: Back Navigation in DeeX Pay

Field	Value
Module	DeeX Pay
Priority	Medium
Type	Functional
Preconditions	On any DeeX Pay sub-page
Test Steps	1. Tap back button
Expected Result	Returns to previous step or home
Actual Result	_[Leave blank]_
Status	 Pending

Virtual Cards Test Cases

TC-VC-001: Card List Display

Field	Value
Module	Virtual Cards
Priority	High
Type	UI
Preconditions	On /virtual-cards
Test Steps	1. Verify card list
Expected Result	Shopping Card (\$245.80) and Subscriptions (\$52.10, Frozen) displayed
Actual Result	_[Leave blank]_
Status	 Pending

TC-VC-002: KYC Gate Check

Field	Value
Module	Virtual Cards
Priority	High
Type	Functional
Preconditions	Mock kycLevel = 1 (below required Level 2)
Test Steps	1. Verify KYC warning banner shown
Expected Result	Warning "KYC Level 2 Required" with link to complete KYC
Actual Result	_[Leave blank]_
Status	 Pending

TC-VC-003: Card Creation

Field	Value
Module	Virtual Cards
Priority	High
Type	Functional
Preconditions	On /virtual-cards, kycLevel >= 2
Test Steps	1. Tap "Create New Card (\$2)" 2. Enter label 3. Select funding source 4. Tap "Create Card"

Expected Result	New card added to list with random last4, toast "Virtual card created!"
Actual Result	_[Leave blank]_
Status	 Pending

TC-VC-004: Card Funding

Field	Value
Module	Virtual Cards
Priority	High
Type	Functional
Preconditions	On card detail, tap "Fund"
Test Steps	1. Enter amount 2. Select source wallet 3. Tap "Fund Card"
Expected Result	Toast "funded to card", returned to detail
Actual Result	_[Leave blank]_
Status	 Pending

TC-VC-005: Freeze/Unfreeze Toggle

Field	Value
Module	Virtual Cards
Priority	High
Type	Functional
Preconditions	On card detail, card is active
Test Steps	1. Tap "Freeze"
Expected Result	Card status changes to frozen, overlay appears on card visual, toast "Card frozen"
Actual Result	_[Leave blank]_
Status	 Pending

TC-VC-006: Card Deletion

Field	Value
Module	Virtual Cards
Priority	High

Type	Functional
Preconditions	On card detail
Test Steps	1. Tap "Delete"
Expected Result	Card removed from list, toast "Card deleted", returned to list
Actual Result	_[Leave blank]_
Status	 Pending

TC-VC-007: Card Number Reveal

Field	Value
Module	Virtual Cards
Priority	High
Type	Functional
Preconditions	On card detail
Test Steps	1. Tap eye icon
Expected Result	Full card number, expiry (03/29), and CVV (412) revealed
Actual Result	_[Leave blank]_
Status	 Pending

TC-VC-008: Spending Limits Edit

Field	Value
Module	Virtual Cards
Priority	Medium
Type	Functional
Preconditions	On card detail, tap "Limits"
Test Steps	1. Edit daily and monthly limits 2. Tap "Save Limits"
Expected Result	Toast "Limits updated", returned to detail
Actual Result	_[Leave blank]_
Status	 Pending

TC-VC-009: Transaction List

Field	Value
Module	Virtual Cards
Priority	Medium
Type	UI
Preconditions	On card detail with transactions
Test Steps	1. Scroll to transactions section
Expected Result	Transaction list with description, amount, date
Actual Result	_[Leave blank]_
Status	 Pending

TC-VC-010: Empty State

Field	Value
Module	Virtual Cards
Priority	Low
Type	UI
Preconditions	No cards exist
Test Steps	1. Verify empty state
Expected Result	"No virtual cards yet" message with icon
Actual Result	_[Leave blank]_
Status	 Pending

Rewards Test Cases

TC-REW-001: Points Balance Display

Field	Value
Module	Rewards
Priority	High
Type	UI
Preconditions	On /rewards
Test Steps	1. Verify DeeXPoints card
Expected Result	Shows 2,450 points ≈ ₦24,500

Actual Result	_[Leave blank]_
Status	Pending

TC-REW-002: Redeem Flow

Field	Value
Module	Rewards
Priority	High
Type	Functional
Preconditions	On /rewards, have ≥ 500 points
Test Steps	1. Tap DeeXPoints card 2. Enter 500 points 3. Tap "Continue"
Expected Result	Confirmation page with points, rate, cash value, destination bank
Actual Result	_[Leave blank]_
Status	Pending

TC-REW-003: Minimum Redemption Validation

Field	Value
Module	Rewards
Priority	High
Type	Functional
Preconditions	On redeem page
Test Steps	1. Enter 100 points
Expected Result	Error "Minimum 500 points (₦5,000)" shown, Continue button disabled
Actual Result	_[Leave blank]_
Status	Pending

TC-REW-004: Exceeds Balance Validation

Field	Value
Module	Rewards
Priority	High
Type	Functional

Preconditions	On redeem page, balance = 2,450
Test Steps	1. Enter 3000 points
Expected Result	Error "Exceeds your balance" shown
Actual Result	_[Leave blank]_
Status	 Pending

TC-REW-005: Referral Link Copy

Field	Value
Module	Rewards
Priority	High
Type	Functional
Preconditions	On /rewards
Test Steps	1. Tap copy button on referral link
Expected Result	Link copied, checkmark shown
Actual Result	_[Leave blank]_
Status	 Pending

TC-REW-006: Referral List Display

Field	Value
Module	Rewards
Priority	Medium
Type	UI
Preconditions	Navigate to /referrals
Test Steps	1. Verify referral list
Expected Result	5 referrals shown with status (Traded/Signed up) and earnings
Actual Result	_[Leave blank]_
Status	 Pending

TC-REW-007: Leaderboard Display

Field	Value
Module	Rewards

Priority	Low
Type	UI
Preconditions	On /referrals
Test Steps	1. Scroll to leaderboard
Expected Result	Top 4 referrers shown with rank, name, referrals, earnings
Actual Result	_[Leave blank]_
Status	 Pending

TC-REW-008: Streak Calendar

Field	Value
Module	Rewards
Priority	Medium
Type	UI
Preconditions	On /rewards
Test Steps	1. Verify daily streak calendar
Expected Result	7-day calendar with checkmarks for completed days
Actual Result	_[Leave blank]_
Status	 Pending

TC-REW-009: Earnings Log

Field	Value
Module	Rewards
Priority	Medium
Type	UI
Preconditions	On /rewards → Earnings Log tab
Test Steps	1. Verify earnings entries
Expected Result	11 entries shown with source, category, points, date
Actual Result	_[Leave blank]_
Status	 Pending

TC-REW-010: Redemption History

Field	Value
Module	Rewards
Priority	Medium
Type	UI
Preconditions	On /rewards → Redemptions tab
Test Steps	1. Verify redemption entries
Expected Result	4 entries with points, cash value, status (Approved/Rejected/Processing)
Actual Result	_[Leave blank]_
Status	 Pending

KYC Test Cases

TC-KYC-001: Overview Display

Field	Value
Module	KYC
Priority	High
Type	UI
Preconditions	On /kyc
Test Steps	1. Verify limits summary 2. Verify 3 level cards 3. Verify progress bar
Expected Result	Trading/withdrawal limits shown, 3 levels with requirements, progress 0/3
Actual Result	_[Leave blank]_
Status	 Pending

TC-KYC-002: Level 1 Form

Field	Value
Module	KYC
Priority	High
Type	Functional
Preconditions	On /kyc, tap Level 1

Test Steps	1. Verify BVN input (max 11 chars) 2. Verify liveness check area
Expected Result	BVN field and selfie upload area visible
Actual Result	_[Leave blank]_
Status	 Pending

TC-KYC-003: Level 2 Form

Field	Value
Module	KYC
Priority	High
Type	Functional
Preconditions	On /kyc, tap Level 2
Test Steps	1. Verify government ID upload 2. Verify proof of address upload
Expected Result	Two upload areas for ID and address proof
Actual Result	_[Leave blank]_
Status	 Pending

TC-KYC-004: Level 3 Form

Field	Value
Module	KYC
Priority	High
Type	Functional
Preconditions	On /kyc, tap Level 3
Test Steps	1. Verify employment docs upload 2. Verify income source dropdown 3. Verify risk questionnaire 4. Verify 2FA requirement notice
Expected Result	All Level 3 fields visible with 2FA warning
Actual Result	_[Leave blank]_
Status	 Pending

TC-KYC-005: BVN Validation

Field	Value
Module	KYC
Priority	High
Type	Functional
Preconditions	On Level 1 form
Test Steps	1. Try entering more than 11 characters
Expected Result	Input capped at 11 characters (maxLength={11})
Actual Result	_[Leave blank]_
Status	 Pending

TC-KYC-006: Submission Toast

Field	Value
Module	KYC
Priority	High
Type	Functional
Preconditions	On any KYC level form
Test Steps	1. Tap "Submit"
Expected Result	Toast "KYC Level X submitted for review", returned to overview after 1.5s
Actual Result	_[Leave blank]_
Status	 Pending

TC-KYC-007: Limits Display

Field	Value
Module	KYC
Priority	Medium
Type	UI
Preconditions	On /kyc overview
Test Steps	1. Verify limits card
Expected Result	Trading and withdrawal limits shown based on highest completed level
Actual Result	_[Leave blank]_
Status	 Pending

TC-KYC-008: Upgrade CTA

Field	Value
Module	KYC
Priority	Medium
Type	UI
Preconditions	On /kyc overview, not all levels completed
Test Steps	1. Verify upgrade CTA card
Expected Result	"Ready for the highest tier?" card with "Upgrade to Level X" button
Actual Result	_[Leave blank]_
Status	Pending

Admin Test Cases

TC-ADM-001: Admin Dashboard Loads

Field	Value
Module	Admin
Priority	High
Type	Functional
Preconditions	Navigate to /admin
Test Steps	1. Verify dashboard loads 2. Verify metrics cards 3. Verify performance chart
Expected Result	4 metric cards (total payout, active users, customers, merchants), performance area chart, quick links
Actual Result	_[Leave blank]_
Status	Pending

TC-ADM-002: Balance Toggle in Admin

Field	Value
Module	Admin
Priority	Medium
Type	Functional

Preconditions	On admin dashboard
Test Steps	1. Tap "SHOW BALANCE"
Expected Result	Metrics values revealed from ""**"
Actual Result	_[Leave blank]_
Status	 Pending

TC-ADM-003: KYC Approve/Reject

Field	Value
Module	Admin
Priority	High
Type	Functional
Preconditions	On Admin → KYC, pending submission visible
Test Steps	1. Click approve button on pending KYC
Expected Result	Toast "KYC approved — limits updated"
Actual Result	_[Leave blank]_
Status	 Pending

TC-ADM-004: KYC Detail Dialog

Field	Value
Module	Admin
Priority	High
Type	Functional
Preconditions	On Admin → KYC
Test Steps	1. Click eye icon on any KYC entry
Expected Result	Dialog opens with info grid, limits preview, requirements checklist, documents, actions
Actual Result	_[Leave blank]_
Status	 Pending

TC-ADM-005: Compliance Alert List

Field	Value
Module	Admin

Priority	High
Type	UI
Preconditions	On Admin → Compliance
Test Steps	1. Verify alert stats cards 2. Verify alert table
Expected Result	4 stat cards (total, critical, pending, auto-suspended), alert table with severity, user, trigger, status
Actual Result	_[Leave blank]_
Status	☐ Pending

TC-ADM-006: Compliance Alert Detail

Field	Value
Module	Admin
Priority	High
Type	Functional
Preconditions	On compliance alerts, click an alert row
Test Steps	1. Verify detail view
Expected Result	User info, trigger description, activity details grid, event timeline, admin actions panel, notes textarea
Actual Result	_[Leave blank]_
Status	☐ Pending

TC-ADM-007: Compliance Rule Toggle

Field	Value
Module	Admin
Priority	High
Type	Functional
Preconditions	On Compliance → Rules Engine
Test Steps	1. Toggle a rule switch
Expected Result	Rule enabled/disabled, toast confirmation, rule card opacity changes
Actual Result	_[Leave blank]_
Status	☐ Pending

TC-ADM-008: Gift Card Order Approval

Field	Value
Module	Admin
Priority	High
Type	Functional
Preconditions	On Admin → Gift Cards, pending order visible
Test Steps	1. Click approve on pending order
Expected Result	Toast "Approved! ₦X credited to User.", order status changes to approved
Actual Result	_[Leave blank]_
Status	 Pending

TC-ADM-009: Gift Card Brand Management

Field	Value
Module	Admin
Priority	High
Type	Functional
Preconditions	On Admin → Gift Cards → Brands & Rates
Test Steps	1. Click "Add Brand" 2. Fill form 3. Submit
Expected Result	New brand added to list, toast confirmation
Actual Result	_[Leave blank]_
Status	 Pending

TC-ADM-010: User Search

Field	Value
Module	Admin
Priority	High
Type	Functional
Preconditions	On Admin → Users
Test Steps	1. Enter "Divine" in search
Expected Result	Table filtered to show only Divine Omajuwa
Actual Result	_[Leave blank]_

Status	 Pending
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TC-ADM-011: User Status Filter

Field	Value
Module	Admin
Priority	High
Type	Functional
Preconditions	On Admin → Users
Test Steps	1. Tap "Flagged" filter
Expected Result	Only flagged users shown (Victor Odigili)
Actual Result	_[Leave blank]_
Status	 Pending

TC-ADM-012: Order Status Filter

Field	Value
Module	Admin
Priority	High
Type	Functional
Preconditions	On Admin → Orders
Test Steps	1. Tap "FAILED" filter
Expected Result	Only failed orders shown with retry buttons
Actual Result	_[Leave blank]_
Status	 Pending

TC-ADM-013: Auto-Pay Toggle

Field	Value
Module	Admin
Priority	High
Type	Functional
Preconditions	On Admin → Orders
Test Steps	1. Toggle AUTO PAY switch

Expected Result	Toast "Auto Pay enabled/disabled"
Actual Result	_[Leave blank]_
Status	 Pending

TC-ADM-014: Payout Retry

Field	Value
Module	Admin
Priority	High
Type	Functional
Preconditions	On Admin → Orders or Payouts, failed item visible
Test Steps	1. Click "Retry" on failed payout
Expected Result	Toast "Retrying payout for User..."
Actual Result	_[Leave blank]_
Status	 Pending

TC-ADM-015: Wallet Tab Switching

Field	Value
Module	Admin
Priority	High
Type	Functional
Preconditions	On Admin → Wallets
Test Steps	1. Toggle between "DeeX Wallet" and "Customers Wallet"
Expected Result	Balance cards, asset cards, and activity table update accordingly
Actual Result	_[Leave blank]_
Status	 Pending

TC-ADM-016: Auto-Swap Toggle

Field	Value
Module	Admin
Priority	Medium
Type	Functional

Preconditions	On Admin → Wallets
Test Steps	1. Toggle AUTO SWAP switch
Expected Result	Toast "Auto Swap enabled/disabled"
Actual Result	_[Leave blank]_
Status	 Pending

TC-ADM-017: Audit Log Display

Field	Value
Module	Admin
Priority	Medium
Type	UI
Preconditions	On Admin → Audit Log
Test Steps	1. Verify audit log table
Expected Result	8 entries showing admin, action, target, details, date
Actual Result	_[Leave blank]_
Status	 Pending

TC-ADM-018: Report Navigation

Field	Value
Module	Admin
Priority	Medium
Type	Functional
Preconditions	On Admin → Reports
Test Steps	1. Click "Revenue" report card
Expected Result	Bar chart and metrics (This Month, Last Month, Total) shown
Actual Result	_[Leave blank]_
Status	 Pending

TC-ADM-019: Fee Configuration Display

Field	Value
Module	Admin

Priority	Medium
Type	UI
Preconditions	On Admin → Settings → Fees
Test Steps	1. Verify fee list
Expected Result	Crypto Trade Fee (1.0%), Gift Card Fee (2.5%), Withdrawal Fee (₦50), DeeX Pay Fee (0.5%)
Actual Result	_[Leave blank]_
Status	 Pending

TC-ADM-020: Pagination Works

Field	Value
Module	Admin
Priority	Medium
Type	Functional
Preconditions	On any admin table with > 10 items
Test Steps	1. Click "Next" or page number
Expected Result	Table updates to show next page, "Showing X–Y of Z" updates
Actual Result	_[Leave blank]_
Status	 Pending

Security Test Cases

TC-SEC-001: PIN Brute Force Protection

Field	Value
Module	Security
Priority	High
Type	Security
Preconditions	On /pin screen
Test Steps	1. Enter wrong PIN 10 times in a row
Expected Result	No lockout implemented (gap — should implement rate limiting)
Actual Result	_[Leave blank]_
Status	 Pending

TC-SEC-002: Session Termination Notification

Field	Value
Module	Security
Priority	High
Type	Functional
Preconditions	Logged in from new device
Test Steps	1. Check notifications
Expected Result	"Session Terminated" notification visible
Actual Result	_[Leave blank]_
Status	Pending

TC-SEC-003: 2FA Toggle

Field	Value
Module	Security
Priority	High
Type	Functional
Preconditions	On /security
Test Steps	1. Toggle "Two-Factor Auth"
Expected Result	Toggle state changes (no actual 2FA setup flow — gap)
Actual Result	_[Leave blank]_
Status	Pending

TC-SEC-004: Biometric Toggle

Field	Value
Module	Security
Priority	Medium
Type	Functional
Preconditions	On /security
Test Steps	1. Toggle "Biometric Login"
Expected Result	Toggle state changes (no actual biometric integration — gap)

Actual Result	_[Leave blank]_
Status	 Pending

TC-SEC-005: Login History Display

Field	Value
Module	Security
Priority	Medium
Type	UI
Preconditions	On /security
Test Steps	1. Scroll to Login History
Expected Result	4 entries with device, date, time, success/failure status
Actual Result	_[Leave blank]_
Status	 Pending

TC-SEC-006: Active Session Termination

Field	Value
Module	Security
Priority	High
Type	Functional
Preconditions	On /security, Active Sessions section
Test Steps	1. Click terminate icon on non-current session
Expected Result	Session removed from list (no actual backend call — gap)
Actual Result	_[Leave blank]_
Status	 Pending

TC-SEC-007: Sensitive Data Masking

Field	Value
Module	Security
Priority	High
Type	Security
Preconditions	On any page with balance display

Test Steps	1. Toggle balance visibility off
Expected Result	All monetary values masked with "....."
Actual Result	_[Leave blank]_
Status	 Pending

TC-SEC-008: Card Number Masking

Field	Value
Module	Security
Priority	High
Type	Security
Preconditions	On virtual card detail
Test Steps	1. Verify default card number display
Expected Result	Card number masked as ".... XXXX" by default
Actual Result	_[Leave blank]_
Status	 Pending

Navigation & Layout Test Cases

TC-NAV-001: Bottom Navigation

Field	Value
Module	Navigation
Priority	High
Type	Functional
Preconditions	On any page with bottom nav
Test Steps	1. Tap Home, Wallet, +, Rewards, Activity
Expected Result	Each tab navigates to correct page, active tab highlighted
Actual Result	_[Leave blank]_
Status	 Pending

TC-NAV-002: Center Action Button

Field	Value
Module	Navigation
Priority	High
Type	Functional
Preconditions	Bottom nav visible
Test Steps	1. Tap center "+" button
Expected Result	Navigated to /quick-action
Actual Result	_[Leave blank]_
Status	Pending

TC-NAV-003: Mobile Layout Constraint

Field	Value
Module	Navigation
Priority	Medium
Type	UI
Preconditions	View app on desktop browser
Test Steps	1. Verify layout
Expected Result	Content constrained to max-width 430px, centered on screen
Actual Result	_[Leave blank]_
Status	Pending

TC-NAV-004: 404 Page

Field	Value
Module	Navigation
Priority	Low
Type	Functional
Preconditions	Navigate to unknown route
Test Steps	1. Navigate to /nonexistent
Expected Result	NotFound page displayed
Actual Result	_[Leave blank]_
Status	Pending

Regression Test Scenarios

#	Scenario	Steps	Expected
R-001	Full user journey	Onboarding → Login → PIN → Dashboard → Sell Crypto → Receipt	All pages load and navigate correctly
R-002	Gift card trade flow	Dashboard → Gift Cards → Select brand → Enter value → Upload → Submit → Status	Order saved to localStorage
R-003	Admin KYC workflow	Admin → KYC → View pending → Approve → Verify toast	KYC approved, limits updated message
R-004	Compliance alert handling	Admin → Compliance → View alert → Detail → Approve	Alert status updated
R-005	Reward redemption	Rewards → Redeem → Enter 500 → Confirm → Success	Redemption submitted for approval
R-006	Virtual card lifecycle	Create → Fund → Freeze → Unfreeze → Delete	All operations succeed with toasts
R-007	Bill payment flow	Bills → Airtime → MTN → Enter details → Pay → Success	Payment successful page shown
R-008	DeeX Pay merchant payment	DeeX Pay → Pay Merchant → Enter ID → Select crypto → Pay	Payment sent success page
R-009	Admin gift card management	Admin → Gift Cards → Approve order → Toggle brand → Add brand	All operations succeed
R-010	Cross-module navigation	Dashboard → Wallet → Swap → Activity → Profile → Security → Back	All back buttons work correctly

4. System Architecture

DeeX — System Architecture Overview

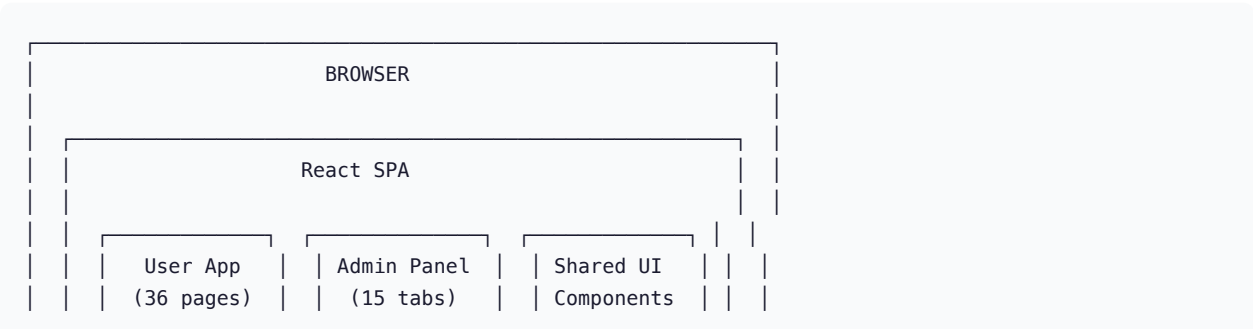
1. System Classification

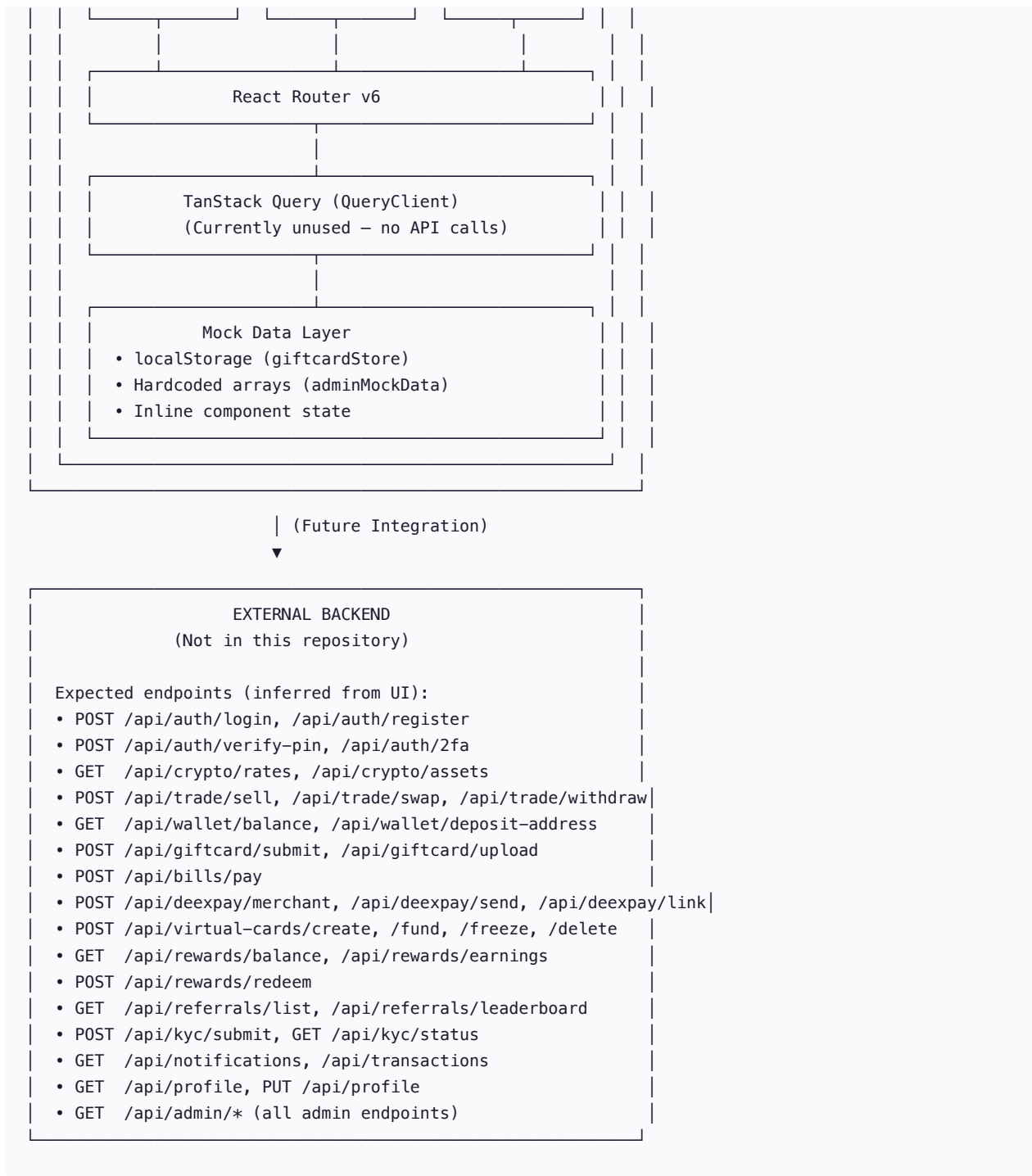
Type: Frontend Prototype (Single-Page Application)
Backend Status: Exists separately (not in this repository)
Data Layer: Fully mocked — localStorage + hardcoded arrays
Deployment: Lovable platform (Vercel-like hosting)

2. Technology Stack

Layer	Technology	Version
Framework	React	18.3.1
Build Tool	Vite	5.4.19
Language	TypeScript	5.8.3
Routing	React Router DOM	6.30.1
Styling	Tailwind CSS	3.4.17
UI Components	shadcn-ui + Radix UI	Various
State Management	TanStack Query (client)	5.83.0
Forms	React Hook Form + Zod	7.61.1 / 3.25.76
Charts	Recharts	2.15.4
Animations	Framer Motion	12.35.1
Notifications	Sonner + Toast	1.7.4
Testing	Vitest + Testing Library	3.2.4 / 16.0.0
Linting	ESLint (typescript-eslint)	9.32.0

3. Architecture Diagram





4. Folder Structure

```
src/
├─ App.tsx           # Root component with all routes
├─ main.tsx          # Entry point
├─ index.css          # Global styles + Tailwind
├─ vite-env.d.ts      # Vite type declarations
├─
├─ pages/             # 36 page components
│   └─ Onboarding.tsx # 3-slide onboarding flow
│   └─ Login.tsx      # Email/password + social login
```

- └─ SignUp.tsx # Registration form
- └─ PinLock.tsx # 4-digit PIN entry
- └─ Dashboard.tsx # Main dashboard with rates, actions, transactions
- └─ Wallet.tsx # Asset list with swipeable balance card
- └─ Deposit.tsx # Crypto deposit (QR + address)
- └─ SellCrypto.tsx # Sell flow (wallet or external source)
- └─ SwapCrypto.tsx # Crypto-to-crypto swap
- └─ WithdrawCrypto.tsx # Withdraw to external wallet
- └─ SendMoney.tsx # Send NGN to bank account
- └─ AssetDetail.tsx # Individual asset page
- └─ Receipt.tsx # Transaction receipt
- └─ TransactionDetail.tsx # Transaction details
- └─ DeexPay.tsx # Pay merchant, send cash, generate link
- └─ GiftCards.tsx # Sell gift cards (multi-step flow)
- └─ BillPayment.tsx # Airtime, data, electricity, betting
- └─ Activity.tsx # Transaction history with filters
- └─ Rewards.tsx # Points, redemption, streak, referrals
- └─ ReferralDashboard.tsx # Referral stats and leaderboard
- └─ Notifications.tsx # Notification inbox
- └─ Profile.tsx # Profile overview and settings links
- └─ EditProfile.tsx # Edit profile form
- └─ GenerateStatement.tsx # Account statement generator
- └─ AboutDeeX.tsx # About, terms, privacy, AML
- └─ SecuritySettings.tsx # PIN, biometric, 2FA, sessions
- └─ BankAccounts.tsx # Manage linked bank accounts
- └─ KycVerification.tsx # 3-tier KYC submission
- └─ Support.tsx # Live chat, email, WhatsApp, FAQ
- └─ QuickAction.tsx # Quick action menu
- └─ VirtualCards.tsx # Visa virtual card management
- └─ TradeStreak.tsx # Trade streak tracking
- └─ AdminPanel.tsx # Admin dashboard shell
- └─ AdminUserDetail.tsx # Individual user detail view
- └─ NotFound.tsx # 404 page

└─ components/

- └─ ui/ # 48 shadcn-ui components
 - └─ button.tsx, input.tsx, dialog.tsx, etc.
 - └─ table.tsx, chart.tsx, form.tsx, etc.
- └─ layout/
 - └─ MobileLayout.tsx # 430px max-width container
 - └─ BottomNav.tsx # 5-tab bottom navigation
- └─ admin/ # 15 admin sub-components
 - └─ AdminLayout.tsx # Sidebar + header layout
 - └─ AdminDashboard.tsx # Metrics + charts
 - └─ AdminWallets.tsx # Wallet management
 - └─ AdminOrders.tsx # Order management
 - └─ AdminPayouts.tsx # Payout tracking
 - └─ AdminUsers.tsx # User management
 - └─ AdminKYC.tsx # KYC approval workflow
 - └─ AdminCompliance.tsx # Fraud detection + rules
 - └─ AdminVirtualCards.tsx
 - └─ AdminGiftCards.tsx # Brand + order management
 - └─ AdminBillPayments.tsx
 - └─ AdminAuditLog.tsx
 - └─ AdminRewards.tsx # Rewards config + influencers
 - └─ AdminReports.tsx # Analytics reports
 - └─ AdminSettings.tsx # Fees, payroll, security
 - └─ AdminUtils.tsx # Shared badges, dialogs, pagination
 - └─ ResponsiveTable.tsx
- └─ rewards/

```

├── ──┬─ SignUpBonus.tsx
│     ──┬─ CashbackCard.tsx
│       ──┬─ EarningsLog.tsx
│         ──┬─ RedemptionHistory.tsx
│           ──┬─ CryptoIcon.tsx          # Crypto asset icons
│             ──┬─ ProviderIcon.tsx      # Provider/telco icons
│               ──┬─ NewBadge.tsx        # "NEW" badge component
│                 ──┬─ NavLink.tsx       # Navigation link
│                   ──┬─ PageTransition.tsx # Framer Motion transitions
│                     ──┬─ EmptyState.tsx # Empty state component
│                       ──┬─ SkeletonLoader.tsx # Loading skeleton
│
├── ──┬─ data/                          # Mock data layer
│     ──┬─ adminMockData.ts            # All admin mock data (656 lines)
│       ──┬─ giftcardData.ts           # Gift card brands, rates, localStorage store
│
├── ──┬─ hooks/
│     ──┬─ use-toast.ts                # Toast hook
│       ──┬─ use-mobile.tsx           # Mobile detection hook
│
├── ──┬─ lib/
│     ──┬─ utils.ts                    # cn() utility (clsx + twMerge)
│
├── ──┬─ test/
│     ──┬─ setup.ts                    # Vitest setup
│       ──┬─ example.test.ts           # Example test

```

5. Routing Architecture

All routes are client-side via React Router v6. No server-side routing.

Route	Component	Layout	Auth Required
`/`	Navigate to `/onboarding`	None	No
`/onboarding`	Onboarding	MobileLayout (hideNav)	No
`/login`	Login	MobileLayout (hideNav)	No
`/signup`	SignUp	MobileLayout (hideNav)	No
`/pin`	PinLock	MobileLayout (hideNav)	No
`/dashboard`	Dashboard	MobileLayout + BottomNav	Yes (mocked)
`/wallet`	Wallet	MobileLayout + BottomNav	Yes
`/deposit`	Deposit	MobileLayout (hideNav)	Yes
`/sell-crypto`	SellCrypto	MobileLayout (hideNav)	Yes
`/swap-crypto`	SwapCrypto	MobileLayout (hideNav)	Yes
`/withdraw`	WithdrawCrypto	MobileLayout (hideNav)	Yes
`/send-money`	SendMoney	MobileLayout (hideNav)	Yes
`/asset/:symbol`	AssetDetail	MobileLayout (hideNav)	Yes

`/receipt`	Receipt	MobileLayout (hideNav)	Yes
`/transaction-detail`	TransactionDetail	MobileLayout (hideNav)	Yes
`/deex-pay`	DeexPay	MobileLayout (hideNav)	Yes
`/giftcards`	GiftCards	MobileLayout (hideNav)	Yes
`/bills/:type`	BillPayment	MobileLayout (hideNav)	Yes
`/activity`	Activity	MobileLayout + BottomNav	Yes
`/rewards`	Rewards	MobileLayout + BottomNav	Yes
`/referrals`	ReferralDashboard	MobileLayout (hideNav)	Yes
`/notifications`	Notifications	MobileLayout (hideNav)	Yes
`/profile`	Profile	MobileLayout (hideNav)	Yes
`/edit-profile`	EditProfile	MobileLayout (hideNav)	Yes
`/generate-statement`	GenerateStatement	MobileLayout (hideNav)	Yes
`/about`	AboutDeeX	MobileLayout (hideNav)	Yes
`/security`	SecuritySettings	MobileLayout (hideNav)	Yes
`/bank-accounts`	BankAccounts	MobileLayout (hideNav)	Yes
`/kyc`	KycVerification	MobileLayout (hideNav)	Yes
`/support`	Support	MobileLayout (hideNav)	Yes
`/quick-action`	QuickAction	MobileLayout (hideNav)	Yes
`/virtual-cards`	VirtualCards	MobileLayout (hideNav)	Yes
`/trade-streak`	TradeStreak	MobileLayout (hideNav)	Yes
`/admin`	AdminPanel	AdminLayout	Yes (admin)
`/admin/users/:id`	AdminUserDetail	AdminLayout	Yes (admin)
`*`	NotFound	MobileLayout	No

Note: No route guards exist. All routes are publicly accessible. Authentication is purely visual — no token checking, no protected routes.

6. State Management

Pattern	Usage	Scope
`useState`	All form inputs, UI toggles, step navigation	Component-level
`localStorage`	Gift card orders/brands, onboarding flag	Persistent
`TanStack Query`	QueryClient initialized but not used	N/A
No global state	No Context API, no Redux, no Zustand	—

Gap: No global auth state, no user session management, no centralized data store. Each page manages its own state independently.

7. Data Flow



No data persists across sessions except:

- `deex\_onboarded` flag
- `deex\_gc\_brands` (gift card brands)
- `deex\_gc\_orders` (gift card orders)

8. External Integrations (Expected)

Integration	Purpose	Status
Crypto Exchange API (Obiex)	Crypto rates, deposits, swaps	✗ Not integrated
Payment Gateway (Hizo)	Payout processing	✗ Not integrated
Bank API (Palmpay, Glyde)	NGN wallet balances	✗ Not integrated
BVN Verification	KYC Level 1 identity check	✗ Not integrated
SMS/Email Service	OTP, notifications	✗ Not integrated
Push Notifications	Real-time alerts	✗ Not integrated
QR Scanner	Merchant payment, deposit	✗ Not integrated
Camera/Media	KYC document upload, selfie	✗ Not integrated
Biometric API	Fingerprint/Face ID	✗ Not integrated
WhatsApp API	Support chat	✓ Link only (wa.me)

9. Security Architecture (Current)

Feature	Implementation	Status
PIN Lock	Hardcoded "1234" in component	⚠️ Insecure
Biometric	UI button only, no WebAuthn	❌ Missing
2FA	Toggle only, no TOTP setup	❌ Missing
Session Management	None (no tokens)	❌ Missing
Route Guards	None	❌ Missing
Input Validation	Minimal (only BVN maxLength)	⚠️ Weak
XSS Protection	React default escaping	✅ Basic
CSRF Protection	N/A (no API calls)	N/A
HTTPS	Via Lovable hosting	✅

10. Performance Considerations

Aspect	Current State	Notes
Bundle Size	Moderate (~495 packages)	Could benefit from code splitting
Route-level splitting	None	All pages bundled together
Image optimization	Placeholder SVGs only	No real images
Lazy loading	None	All components eagerly loaded
Memoization	`useMemo` in GiftCards only	Minimal optimization
Virtual scrolling	None	Long lists render all items

11. Testing Setup

Tool	Configuration	Coverage
Vitest	jsdom environment	1 example test
Testing Library	@testing-library/react	Not used in any real tests
Coverage	Not configured	0%

12. Deployment Architecture

```

Lovable Platform
├──
├── Build: npm run build (Vite)
├── Output: dist/ (static files)

```

- └─ Hosting: Lovable CDN
- └─ Custom Domain: Supported via Lovable settings

No CI/CD pipeline configured. Changes pushed via git are auto-deployed by Lovable.

5. Gap Analysis

DeeX — Gap Analysis

Critical Gaps

GAP-001: No Authentication Backend

Severity: Critical

Module: Authentication

Description: Login, signup, and PIN lock have zero backend integration. No user accounts are created, no passwords are validated, no sessions are managed. The PIN is hardcoded as "1234" in the component.

Impact: The entire app is publicly accessible with no real security. Any user can access any page by typing the URL.

Recommendation: Integrate with the existing backend auth API. Implement JWT-based session management, protected routes, and proper password hashing.

GAP-002: No API Integration Anywhere

Severity: Critical

Module: All

Description: Every feature uses hardcoded mock data. No API calls are made anywhere in the codebase. TanStack Query is initialized but never used.

Impact: The app is purely a visual prototype. No real transactions, balances, or user data exist.

Recommendation: Create API service layer and replace all mock data with real API calls. Use TanStack Query for data fetching and caching.

GAP-003: No Route Protection

Severity: Critical

Module: Navigation

Description: All routes are publicly accessible. There are no route guards, auth checks, or role-based access controls.

Impact: Anyone can access the admin panel at `/admin` without authentication.

Recommendation: Implement route guards that check auth tokens and user roles before rendering protected pages.

GAP-004: Hardcoded PIN

Severity: Critical

Module: Authentication

Description: The PIN lock screen uses a hardcoded `correctPin = "1234"` and even displays "Default PIN: 1234" as a hint.

Impact: Zero security. The hint text should never be in production.

Recommendation: Remove hint text. Integrate PIN verification with backend. Implement rate limiting after failed attempts.

GAP-005: No Input Validation on Forms

Severity: Critical

Module: All forms

Description: Signup, login, sell crypto, bill payments, and most forms have no validation. Users can submit empty forms, invalid emails, negative amounts, etc.

Impact: Poor UX and potential data integrity issues when backend is connected.

Recommendation: Add Zod validation schemas to all forms. Use React Hook Form's validation features consistently.

GAP-006: No Error Handling

Severity: Critical

Module: All

Description: No try/catch blocks, no error boundaries, no loading states for async operations, no error UI.

Impact: If any API call fails in production, the app will crash or hang silently.

Recommendation: Add error boundaries, loading skeletons, and error toast notifications throughout.

Important Gaps

GAP-007: Forgot Password Not Implemented

Severity: Important

Module: Authentication

Description: The "Forgot Password?" link exists on the login page but navigates nowhere.

Impact: Users who forget their password are locked out permanently.

Recommendation: Implement email OTP or reset link flow.

GAP-008: Social Login Non-Functional

Severity: Important

Module: Authentication

Description: Google and Apple login buttons render but do nothing when clicked.

Impact: Users expecting social login will be confused.

Recommendation: Integrate OAuth providers or remove buttons until ready.

GAP-009: No Real-Time Crypto Rates

Severity: Important

Module: Crypto Trading

Description: Crypto rates are hardcoded in the Dashboard component (BTC ₦97,450,000, etc.).

Impact: Users see stale prices that don't reflect market reality.

Recommendation: Connect to a crypto price API (CoinGecko, Binance) or the backend rates endpoint.

GAP-010: No Transaction Persistence

Severity: Important

Module: Activity, Wallet

Description: All transactions are hardcoded arrays. No transaction history is stored or retrieved.

Impact: Users cannot see their actual transaction history.

Recommendation: Fetch transactions from backend API. Store locally as cache.

GAP-011: Bill Payment Not Connected to Providers

Severity: Important

Module: Bill Payments

Description: Bill payments show success but don't actually top up airtime, pay electricity, etc.

Impact: Users would lose money if this went live without integration.

Recommendation: Integrate with bill payment APIs (e.g., Clubconnect, Vtu.ng, or similar Nigerian providers).

GAP-012: No Real QR Code Generation

Severity: Important

Module: Deposit, DeeX Pay

Description: QR codes are placeholder icons, not actual scannable QR codes.

Impact: Users cannot scan to get deposit addresses or pay merchants.

Recommendation: Use a QR code generation library (qrcode.react) to generate real QR codes from wallet addresses.

GAP-013: No Camera/Upload Integration

Severity: Important

Module: KYC, Gift Cards

Description: KYC document upload and gift card image upload are placeholder divs with no actual file input or camera access.

Impact: Users cannot submit KYC documents or gift card images.

Recommendation: Implement file input with image preview and upload to backend storage (S3, Cloudinary).

GAP-014: Admin Panel Has No Auth

Severity: Important

Module: Admin Dashboard

Description: The admin panel at `/admin` is accessible to anyone who knows the URL.

Impact: Complete security breach — anyone can view and "manage" all user data.

Recommendation: Add admin role check and protect the route.

GAP-015: No Real Notifications System

Severity: Important

Module: Notifications

Description: Notifications are hardcoded static data. No push notifications, no real-time updates.

Impact: Users miss important security alerts and transaction updates.

Recommendation: Implement WebSocket or polling for real-time notifications. Integrate with push notification service.

GAP-016: Duplicate `/admin` Route

Severity: Important

Module: Routing

Description: The `/admin` route is defined twice in App.tsx (lines 85-86).

Impact: Potential routing confusion, though React Router uses the first match.

Recommendation: Remove the duplicate route definition.

GAP-017: No Pagination on User-Facing Lists

Severity: Important

Module: Activity, Wallet

Description: Transaction lists and asset lists render all items at once with no pagination or virtualization.

Impact: Performance degradation with large datasets.

Recommendation: Implement pagination or infinite scroll for transaction history.

Nice-to-Have Gaps

GAP-018: No Dark Mode Toggle

Severity: Nice-to-Have

Module: Settings

Description: next-themes is installed but no dark mode toggle exists in the UI.

Impact: Users cannot switch themes.

Recommendation: Add theme toggle in settings using next-themes.

GAP-019: No Search in User-Facing Pages

Severity: Nice-to-Have

Module: Activity

Description: Activity page has a search input but it only searches by hash ID, not by type or date.

Impact: Limited search capability.

Recommendation: Expand search to include transaction type, amount range, and date filters.

GAP-020: No Offline Support

Severity: Nice-to-Have

Module: All

Description: No service worker, no PWA manifest, no offline fallback.

Impact: App doesn't work without internet.

Recommendation: Add PWA support with service worker for offline balance viewing.

GAP-021: No Accessibility Features

Severity: Nice-to-Have

Module: All

Description: No ARIA labels, no keyboard navigation, no screen reader support.

Impact: Not accessible to users with disabilities.

Recommendation: Add ARIA labels, focus management, and keyboard navigation.

GAP-022: No Internationalization

Severity: Nice-to-Have

Module: All

Description: All text is hardcoded in English. No i18n framework.

Impact: Cannot support multiple languages.

Recommendation: Integrate react-i18next for future localization.

GAP-023: No Analytics Integration

Severity: Nice-to-Have

Module: All

Description: No Google Analytics, Mixpanel, or any tracking.

Impact: Cannot measure user behavior or conversion funnels.

Recommendation: Add analytics SDK and track key events (signups, trades, redemptions).

GAP-024: No Loading States

Severity: Nice-to-Have

Module: All

Description: No skeleton loaders or spinners anywhere in the app.

Impact: Users see blank screens during data loading (when APIs are connected).

Recommendation: Add skeleton loaders to all pages that fetch data.

Gap Summary by Priority

Priority	Count	Key Areas
Critical	6	Auth, API, Validation, Error Handling
Important	11	Social login, rates, bills, QR, uploads, admin auth
Nice-to-Have	7	Dark mode, search, offline, a11y, i18n, analytics
Total	24	

6. Technical Debt

DeeX — Technical Debt Analysis

1. Structural Issues

TD-001: No API Service Layer

Severity: High

Location: Entire codebase

Description: All data is hardcoded inline or in mock files. There is no abstraction layer for API calls. When the backend is ready, every page will need to be rewritten.

Fix: Create `src/services/` directory with typed API client functions. Use TanStack Query for data fetching.

TD-002: No Global State Management

Severity: High

Location: App.tsx, all pages

Description: User state, auth state, and wallet balances are not managed globally. Each page uses independent `useState`. When backend is connected, data will need to be refetched on every navigation.

Fix: Implement React Context for auth/user state, or use TanStack Query's cache for shared data.

TD-003: Monolithic App.tsx

Severity: Medium

Location: `src/App.tsx`

Description: All 37 routes are defined in a single file. No lazy loading, no route grouping.

Fix: Split routes into logical groups (auth, user, admin). Use `React.lazy()` for code splitting.

TD-004: Duplicate Route Definition

Severity: Low

Location: `src/App.tsx:85-86`

Description: `}>` is defined twice.

Fix: Remove the duplicate line.

2. Repeated Logic

TD-005: Dropdown Pattern Repeated 20+ Times

Severity: Medium

Location: SellCrypto, SwapCrypto, Deposit, DeexPay, SendMoney, VirtualCards

Description: Every dropdown (asset selector, bank selector, network selector) is implemented as inline state + conditional rendering. No reusable Dropdown component.

Fix: Create a reusable `DropDownSelect` component that accepts options and returns selected value.

TD-006: Success Page Pattern Repeated

Severity: Medium

Location: SellCrypto, SwapCrypto, GiftCards, BillPayment, DeexPay, SendMoney

Description: Every flow has its own inline success page with similar structure (icon, title, description, buttons).

Fix: Create a reusable `SuccessPage` component with configurable icon, title, description, and actions.

TD-007: Review Page Pattern Repeated

Severity: Medium

Location: SellCrypto, SwapCrypto, GiftCards, BillPayment, DeexPay, SendMoney

Description: Review/confirmation pages all use the same card-with-rows pattern.

Fix: Create a reusable `ReviewCard` component that accepts key-value pairs.

TD-008: Copy-to-Clipboard Pattern Repeated

Severity: Low

Location: Deposit, SellCrypto, DeexPay, Rewards, VirtualCards

Description: Each page implements its own copy state, timeout, and icon toggle.

Fix: Already partially addressed by `CopyButton` in AdminUtils. Extract for user-facing pages too.

3. Hardcoded Values

TD-009: Hardcoded Crypto Rates

Severity: High

Location: `src/pages/Dashboard.tsx:12-18`, `src/pages/SellCrypto.tsx:9-14`

Description: BTC ₦97,450,000, ETH ₦5,830,000, USDT ₦1,535, etc. are hardcoded.

Fix: Fetch from backend API or external price service.

TD-010: Hardcoded PIN

Severity: Critical

Location: `src/pages/PinLock.tsx:11`, `src/pages/WithdrawCrypto.tsx:47`

Description: `correctPin = "1234"` is hardcoded in two places.

Fix: Remove immediately. Verify PIN via backend API.

TD-011: Hardcoded User Data

Severity: High

Location: `src/pages/Dashboard.tsx:94` ("John"), `src/pages/Profile.tsx:29` ("John Doe")

Description: User name is hardcoded as "John Doe" everywhere.

Fix: Fetch from auth context or backend user profile endpoint.

TD-012: Hardcoded Balances

Severity: High

Location: `src/pages/Dashboard.tsx:100` (\$12,450.80), `src/pages/Wallet.tsx:10-17`

Description: All wallet balances are hardcoded.

Fix: Fetch from wallet balance API.

TD-013: Hardcoded KYC Level

Severity: Medium

Location: `src/pages/VirtualCards.tsx:58`

Description: `const kycLevel = 2` is hardcoded.

Fix: Fetch from user profile API.

TD-014: Hardcoded Daily Limits

Severity: Medium

Location: `src/pages/SendMoney.tsx` (¥50,000), `src/pages/DeexPay.tsx` (¥100,000)

Description: Daily limits differ between Send Money and DeeX Pay with no explanation.

Fix: Fetch limits from backend based on user's KYC level.

TD-015: Hardcoded Referral Link

Severity: Low

Location: `src/pages/Rewards.tsx:57`, `src/pages/ReferralDashboard.tsx:25`

Description: Referral link is hardcoded as `https://deex.app/ref/johndoe`.

Fix: Generate from user's actual referral code.

4. State Management Problems

TD-016: No Auth State

Severity: Critical

Location: Entire app

Description: No auth context, no token storage, no session management. The app has no concept of "logged in" vs "logged out".

Fix: Create AuthContext with login/logout/token management. Add route guards.

TD-017: Form State Not Centralized

Severity: Medium

Location: All form pages

Description: Each page manages its own form state with individual `useState` calls instead of using React Hook Form consistently.

Fix: Use React Hook Form + Zod for all forms (already installed but underutilized).

TD-018: Gift Card Store Uses localStorage Directly

Severity: Medium

Location: `src/data/giftcardData.ts`

Description: The giftcardStore reads/writes localStorage directly without any abstraction for future API migration.

Fix: Create a store interface that can swap between localStorage and API implementations.

5. Scalability Risks

TD-019: No Code Splitting

Severity: Medium

Location: `src/App.tsx`

Description: All 36 pages are bundled into a single JS file. Initial load includes admin components that most users never need.

Fix: Use `React.lazy()` + `Suspense` for route-level code splitting, especially for admin pages.

TD-020: No Virtual Scrolling for Long Lists

Severity: Medium

Location: AdminOrders (50 items), AdminPayouts (45 items), AdminUsers (12 items)

Description: All list items are rendered at once. With real data (thousands of orders), this will cause performance issues.

Fix: Implement virtual scrolling (react-window) or server-side pagination.

TD-021: No Image Optimization

Severity: Low

Location: Gift cards, virtual cards, KYC documents

Description: When real images are added, there's no optimization strategy.

Fix: Use Vite's asset handling with lazy loading and compression.

6. Security Concerns

TD-022: Hardcoded PIN (see TD-010)

Severity: Critical

Fix: Remove and implement backend verification.

TD-023: No Input Sanitization

Severity: High

Location: All form inputs

Description: User inputs are rendered directly without sanitization. While React escapes by default, any `dangerouslySetInnerHTML` usage would be vulnerable.

Fix: Audit all rendering of user input. Add Zod validation schemas.

TD-024: No Rate Limiting

Severity: High
Location: PinLock, all form submissions
Description: No protection against brute force attacks on PIN or form spam.
Fix: Implement client-side rate limiting and rely on backend rate limiting.

TD-025: Clipboard API Without Fallback

Severity: Low
Location: Multiple pages using ``navigator.clipboard.writeText``
Description: Clipboard API may not be available in all browsers or contexts.
Fix: Add fallback copy method using `document.execCommand`.

TD-026: External Links Without noopener

Severity: Low
Location: ``src/pages/Support.tsx:100`` (WhatsApp link)
Description: WhatsApp link has ``rel="noopener noreferrer"`` but other external links may not.
Fix: Audit all external links for security attributes.

7. TypeScript Issues

TD-027: Loose TypeScript Config

Severity: Medium
Location: ``tsconfig.json``
Description: ``noImplicitAny: false`, `strictNullChecks: false`, `noUnusedLocals: false`, `noUnusedParameters: false``.
Fix: Enable strict mode incrementally. Fix type errors as they surface.

TD-028: Missing Type Definitions

Severity: Medium
Location: Multiple pages
Description: Many components use ``any`` implicitly or lack proper prop types.
Fix: Add explicit type definitions for all component props and state.

Tech Debt Summary

Category	Count	Critical	High	Medium	Low
Structural	4	0	2	1	1
Repeated Logic	4	0	0	3	1
Hardcoded Values	7	1	3	2	1
State Management	3	1	0	2	0

Scalability	3	0	0	2	1
Security	5	1	2	0	2
TypeScript	2	0	0	2	0
Total	28	3	7	12	6

7. Roadmap

DeeX — One-Year Roadmap

Q1: Stabilization & Core Fixes (Months 1-3)

Goals

- Connect frontend to existing backend
- Fix critical security gaps
- Establish proper state management and data fetching

Features to Ship

1. Real Authentication Integration

- Connect login/signup to backend auth API
- Implement JWT token management
- Add route guards for protected pages
- Implement forgot password flow
- Remove hardcoded PIN and implement real verification

2. API Service Layer

- Create typed API client with Axios/fetch
- Replace all mock data with real API calls
- Set up TanStack Query for caching and refetching

3. Form Validation

- Add Zod schemas to all forms
- Implement consistent error messages
- Add loading states to all submissions

4. Error Handling

- Add error boundaries
- Implement global error toast system
- Add loading skeletons to all data-fetching pages

Technical Priorities

- Enable strict TypeScript mode
- Set up route-level code splitting
- Add comprehensive input validation
- Implement proper session management
- Set up CI/CD pipeline

Success Metrics

- 100% of pages connected to real APIs
 - Zero hardcoded sensitive values
 - All forms validated with Zod
 - Error rate < 1% in production
-

Q2: Core Feature Expansion (Months 4-6)

Goals

- Complete all missing feature integrations
- Improve real-time capabilities
- Enhance user experience

Features to Ship

1. Real-Time Crypto Rates

- Connect to price feed API
- Implement rate auto-refresh (WebSocket or polling)
- Add rate change indicators

2. Bill Payment Integration

- Connect to Nigerian bill payment providers
- Implement real airtime top-up, data purchase, electricity payment
- Add transaction receipt generation

3. QR Code Generation

- Implement real QR code generation for deposit addresses
- Add QR scanner for merchant payments
- Support deep linking for payment links

4. KYC Document Upload

- Implement file upload with image preview
- Connect to document storage (S3/Cloudinary)
- Add camera capture for mobile devices
- Integrate with BVN verification service

5. Push Notifications

- Implement web push notifications
- Real-time transaction alerts
- Security alerts (login from new device)

6. Social Login

- Google OAuth integration
- Apple Sign-In integration
- Account linking for existing users

Technical Priorities

- Implement WebSocket for real-time updates
- Add service worker for PWA support
- Optimize bundle size (target < 500KB initial load)
- Set up comprehensive test coverage (> 60%)

Success Metrics

- Bill payment success rate > 95%
- KYC approval turnaround < 24 hours
- Push notification delivery rate > 90%
- Social login adoption > 30% of new signups

Q3: Growth & Monetization (Months 7-9)

Goals

- Launch revenue-generating features
- Improve user acquisition and retention
- Add advanced admin capabilities

Features to Ship

1. Virtual Card Issuance

- Connect to card issuing provider (e.g., Stripe Issuing, Marqeta)
- Real card creation, funding, and transactions
- International payment support

2. Referral Program V2

- Real-time referral tracking
- Automated point awards on referral trades
- Influencer program with profit-share payouts
- Referral leaderboard with prizes

3. Advanced Analytics Dashboard

- User cohort analysis
- Revenue per user tracking
- Conversion funnel analysis
- Churn prediction

4. Admin Enhancements

- Bulk user operations
- Advanced search and filtering
- Export to CSV/PDF
- Admin activity monitoring
- Role-based admin permissions

5. Fee Configuration UI

- Real-time fee updates from admin panel
- Dynamic fee calculation based on user tier
- Fee transparency in user-facing flows

6. Statement Generation

- Real PDF statement generation
- Date range filtering
- Email delivery of statements

Technical Priorities

- Implement role-based access control (RBAC)
- Add comprehensive audit logging
- Set up A/B testing framework
- Implement feature flags for gradual rollouts

Success Metrics

- Monthly active users growth > 25% QoQ
 - Virtual card adoption > 15% of KYC Level 2+ users
 - Referral-driven signups > 20% of total
 - Admin task completion time reduced by 50%
-

Q4: Scale & Optimization (Months 10-12)

Goals

- Optimize performance at scale
- Expand market reach
- Prepare for regulatory compliance

Features to Ship

1. Performance Optimization

- Virtual scrolling for all long lists
- Image lazy loading and compression
- Service worker caching strategy
- Route preloading
- Target: < 2s Time to Interactive

2. Internationalization

- Multi-language support (English, Yoruba, Hausa, Igbo, French)
- Currency display options
- Regional rate configurations

3. Advanced Security

- Device fingerprinting
- Behavioral biometrics
- Transaction signing
- Hardware security key support
- SOC 2 compliance preparation

4. API Versioning & Documentation

- OpenAPI/Swagger documentation
- API versioning strategy
- Rate limiting and throttling
- Webhook support for third-party integrations

5. Mobile App Preparation

- Responsive design improvements
- React Native compatibility audit
- Shared component library design
- API contract finalization

6. Compliance & Reporting

- Automated regulatory reporting
- Transaction monitoring for AML
- Audit trail export
- Data retention policies

Technical Priorities

- Achieve 90%+ test coverage
- Implement comprehensive monitoring (Sentry, Datadog)

- Set up disaster recovery procedures
- Database optimization and indexing
- CDN and edge caching strategy

Success Metrics

- Page load time < 2s on 3G
- Test coverage > 90%
- Zero critical security incidents
- 99.9% uptime SLA
- Ready for mobile app development

Roadmap Summary

Quarter	Theme	Key Deliverables	Risk Level
Q1	Stabilization	Auth integration, API layer, validation, error handling	High
Q2	Expansion	Real-time rates, bill payments, KYC upload, push notifications	Medium
Q3	Growth	Virtual cards, referrals V2, analytics, admin enhancements	Medium
Q4	Scale	Performance, i18n, advanced security, compliance	Low

Dependencies & Blockers

1. **Backend API Availability** — Q1 progress depends on backend endpoints being ready
2. **Third-Party Integrations** — Bill payment providers, card issuing, BVN verification require business agreements
3. **Regulatory Approval** — Some features may require CBN or SEC approval
4. **Design Resources** — Mobile app preparation and i18n require design system updates

Resource Requirements

Role	Q1	Q2	Q3	Q4
Frontend Engineers	2	2	3	2
Backend Engineers	2	3	2	2
QA Engineers	1	2	2	2
DevOps	0.5	1	1	1
Product Designer	1	1	1	0.5
Product Manager	1	1	1	1